A course in microeconomic theory

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for Anat

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preface

This book provides an introduction to the content and methods of microeconomic theory. The primary target for this book is a first-year graduate student who is looking for an introduction to microeconomic theory that goes beyond the traditional models of the consumer, the firm, and the market. It could also be used by undergraduate majors who seek an "advanced theory" course following traditional intermediate micro. And, for relatively mathematically sophisticated students, it could be used as a nontraditional intermediate micro course. The book presumes, however, that the reader has survived the standard intermediate microeconomics course, and the student who has not gone through such a course may find it helpful on occasion to consult a traditional intermediate text. There are many excellent books around; four that I have recommended to students are Friedman, *Price Theory* (South-Western Publishing, 1986), Hirshleifer, *Price Theory and Applications* (4th edition, Prentice Hall, 1988), Nicholson, *Microeconomic Theory — Basic Principles and Extensions* (3d edition, Dryden Press, 1985), and Pindyck and Rubinfeld, *Microeconomics* (Macmillan, 1989).

The distinguishing feature of this book is that although it treats the traditional models (in parts I and II) it emphasizes more recent variations, with special emphases on the use of noncooperative game theory to model competitive interactions (part III), transactions in which private information plays a role (part IV), and the theory of firms and other nonmarket institutions (part V).

For the most part, the formal mathematical requirements of this book are mild. The reader will have to be able to do constrained optimization problems with inequality constraints, using Lagrange multipliers and the complementary slackness conditions. A cookbook recipe for this technique and an example are presented in an appendix. The reader can trust that for any problem in this book, with exceptions as noted, any solution to the first-order conditions and the complementary slackness conditions is a global optimum. That is, with a few exceptions, only "convex" problems are considered. The reader is expected to know basic probability theory.

For the sake of completeness, I sometimes delve into "optional" topics that either require a significant amount of mathematics or deal with somewhat advanced and/or esoteric topics. This material is presented both in footnotes and in slightly smaller typesize in the text. For the most part, it is up to the reader to supply the necessary mathematical background. The one exception concerns dynamic programming; a second appendix gives just enough background in the techniques of d.p. to get through its uses in this book. I apologize for any frustration the more technical material may cause. I have tried to make the book comprehensible even if the reader skips all this material.

The level of mathematical sophistication required (as opposed to the number of mathematical tools) rises as the book progresses. I have written this book with a first-year graduate student in mind, and I expect that the tolerence of such students for abstract arguments will rise as their first year of study unfolds.

The general style of the book is informal, and the number of theorems and proofs is relatively low. This book is meant to be a skeleton survey of microeconomic theory or, more precisely, of some of the topics and techniques of microeconomic theory; students and instructors will wish to supplement what is here according to their own tastes and interests. I try at the end of each chapter to indicate where the reader should go for more advanced and complete coverage of the topics under discussion.[1] The text is relatively more complete and formal in part III, the introduction to noncooperative game theory, because there are fewer good advanced treatments of that material other than the original papers.

Two caveats are called for by the style I employ. Despite the very impressive number of pages in this book, the number of ideas per page is low relative to a standard text. I prefer, and I think students prefer, books that are relatively chatty. Be that as it may, *this* book is relatively chatty. (This is my excuse for calling this *A Course in* .... Perhaps it would have been better to have called this *Lectures in* ....) I hope that readers who know some of the material aren't put off by this but can get to what they don't know without much delay. Second, I've tried to write a good textbook, which is not to my mind the same thing as a good reference book. The reader who goes back to this book to find something specific may find the exercise somewhat trying. In particular, many important concepts are discussed only in the context of specific problems, with the reader left to think through the general structure. I think this makes for a good textbook; but it doesn't make for a good reference book. Besides not always giving general formulations of concepts, I never give exhaustive surveys of the

[1] These reading lists are products of convenience, which means that they emphasize my own work and the work of close colleagues; there is certainly a Stanford and then American bias to these lists. They are not intended to be exhaustive; they certainly are not intended to establish precedence. It is inevitable that I have omitted many excellent and important citations, and apologies are hereby tendered to anyone offended.

range of applications that can be found in the literature; I hope that the reader, having digested my one example in one context, will be prepared to study other examples and contexts encountered outside this book. And I have not tried to diversify among topics. Some economy is achieved by using related examples (many of which I draw from the literature on industrial organization), and a book this long should economize whenever and wherever possible.

Owing to the limitations of time (and, less importantly, space), I didn't get to a few topics that I would have liked to have included. These fall primarily in the areas of parts IV and V and include: rational expectations with differential information; search; coordination failures; micromodels of market institutions, including auctions; models of political institutions. If ever there is a second edition, perhaps it will be longer still!

Insofar as a textbook can have a plot or a story or a message, I believe that this book does; this is not simply a compendium of many interesting and somewhat connected models. It is traditional, I believe, to give a précis of the "message" in the preface. But I have instead moved this to the end of chapter 1 in the hope that more readers will see it there than would see it here.

Concerning the ordering of topics, I am greatly undecided on one point (and instructors using this book may wish to undo the decision I settled on). To show applications of game theory as the techniques are developed, it may be sensible to interleave parts III and IV. After finishing chapter 12, go on to the earlier sections of chapter 16. Then take up chapter 17 and, when the game theoretic treatment of signaling is encountered, return to chapter 13. The disadvantage is that this breaks up the "oligopoly story" that runs through part III; my own experience is that the oligopoly story generates sufficient interest for a first pass through the techniques, and then parts IV and V provide good opportunities to review the techniques of part III. More generally, the book is filled with material that various instructors may find peripheral or too detailed for a first course; the course that I have taught, which formed the basis for this book, went from the first page to the last in ten weeks, but covered perhaps a third to a half of what fits between.

Most of the chapters come with a set of problems at the end. The serious student should do as many as possible. There are fewer problems per topic than one normally finds in textbooks, and the problems tend to be longer than most textbook problems because they often develop variations on the basic themes that the text doesn't cover. Variations are left to the problems on the principle that "doing" a model is a much better way to learn than reading about the model. A few of these problems require mathematical skills substantially beyond those needed to read the text of the book; the student who completes all the problems will have done extraordinarily well.

In teaching a course based on this book, I've supplemented the material here with reading in other, more traditional, texts, and (as importantly) with discussion of a few case studies from the Harvard Business School. I have found particularly valuable: *The Oil Tanker Shipping Industry in 1983* (Case #9-384-034) for discussion of perfect competition; *GE vs. Westinghouse in Large Turbine Generators (A), (B), and (C)* (#9-380-128,129,130) for discussion of implicit collusion in a noisy environment; *The Lincoln Electric Company* (#376-028) for discussion of incentives, screening, and nonneoclassical models of the firm; and *The Washington Post (A) and (B)* (#667-076,077) for discussion of internal organization, implicit contracts, transaction cost economics, and so forth.

I am grateful to a number of colleagues (some of whom were anonymous reviewers) and students who made helpful comments and suggestions on drafts of this book.[2] It would be impossible to give a comprehensive list, but Dilip Abreu, Anat Admati, David Baron, Elchanan Ben Porath, Ken Binmore, Don Brown, Drew Fudenberg, Oliver Hart, Bengt Holmstrom, David Levine, Nachum Melumad, Andrew McLennan, Paul Milgrom, Barry Nalebuff, John Pencavel, Stefan Reichelstein, Peter Reiss, John Roberts, Ariel Rubinstein, José Scheinkman, Jean Tirole, Guy Weyns, Jin Whang, Bob Wilson, Mark Wolfson, and especially Alex Benos, Marco Li Calzi, and Janet Spitz deserve special mention. I am also grateful to Virginia Barker, who copy-edited the book with skill and grace, and to Jack Repcheck of the Princeton University Press for his unflagging efforts in the preparation of this book and for his confidence in it, even when confronted with the possibility of yet another unexpected hundred pages.

It goes without saying that most of what is contained here I learned from and with many teachers, colleagues, and students. The list of such individuals is much too long to give here, so I will refrain except to note my obvious debt to Robert Wilson and to thank him and all of the rest of you.

Note: For the addendum to the second and subsequent printings, see pages 817–18.

[2] Many of the reviewers will be unhappy that I didn't more directly address their criticisms. But let me take this opportunity to assure them that I took those criticisms to heart, and I think they all helped make this a far better book.

A course in microeconomic theory

chapter one

An overview

This opening chapter sets out some basic concepts and philosophy for the rest of the book. In particular, we address four questions: What are the basic categories in microeconomic theory? What are the purposes of microeconomic theory? How does one's purpose influence the levels of scope, detail, and emphasis in one's model? How will the development of the theory proceed in this book?

1.1. The basic categories: Actors, behavior, institutions, and equilibrium

Microeconomic theory concerns the behavior of individual economic actors and the aggregation of their actions in different institutional frameworks. This one-sentence description introduces four categories: The individual *actor*, traditionally either a consumer or a firm; the *behavior* of the actor, traditionally utility maximization by consumers and profit maximization by firms; an *institutional framework*, which describes what options the individual actors have and what outcomes they receive as a function of the actions of others, traditionally the price mechanism in an impersonal marketplace; and the mode of analysis for modeling how the various actors' behaviors will aggregate within a given framework, traditionally *equilibrium* analysis.

The actors

In the standard treatment of microeconomics, the two types of actors are the individual *consumer* and the *firm*. We will follow the standard approach by regarding the consumer as an actor. For the firm, we will follow the standard approach for a while by treating the firm as an actor. But firms can also be thought of as institutions within which the behavior of various sorts of constituent consumers (workers, managers, suppliers, customers) are aggregated. From this alternative perspective, what a firm does results from the varying desires and behavior of its constituent consumers, the institutional framework the firm provides, and the equilibrium that is attained among the constituent consumers within that institutional framework. Much recent work in microeconomics has been directed towards treating the firm in this fashion, and we will take up this approach in the final part of the book.

Behavior

In the standard approach, behavior always takes the form of constrained maximization. The actor chooses from some specified set of options, selecting the option that maximizes some objective function. In orthodox theory, consumers have *preferences* that are represented by a *utility function*, and they choose in a way that maximizes their utility subject to a *budget constraint*. Firms, on the other hand, are modeled as *maximizing profits* subject to the constraints imposed by their *technological production possibilities set*.

These models of consumer and firm behavior typically strike people as fairly obnoxious. We don't find consumers strolling down the aisles of supermarkets consulting a utility function to maximize when making their choices, nor do we typically think of business executives being guided completely and solely (or even mainly) by the effect of their decisions on business profits. Nonetheless, we will use the standard model of consumer behavior throughout, and we will use the standard model of the firm for most of the book. It behooves us, then, to say why we think such models are useful.

Three rationales are often given. First, our models don't presume that consumers actively maximize some tangible utility function; the presumption is that consumers act *as if* this is what they do. Hence an important part of the theory of individual behavior concerns *testable restrictions* of the models we use: What behavior, if observed, would clearly falsify our models? If the models are not falsified by our observations, then our models are good positive models — perhaps not descriptive as to *why* things happen, but good in describing *what* happens.

Unhappily, rather a lot of data has been collected, especially experimentally, which falsifies the models we will employ. At this we fall back to our second line of defense by saying that such violations may be minor and not amount to much. That is, the standard models may be good approximations of individual behavior, and the conclusions of models built from such approximations may thus be approximately valid. This requires a leap of faith, but it is still a leap that has some intuitive appeal. In fact, we can take this a step further: In many (but not all) of our models, the behavior of the individual is unimportant; instead the aggregate behavior of many individuals matters. If we believe that violations of our models tend to exhibit no particular biases and cancel out at the level of aggregate behavior, then the models may work well. If all we care about are the implications of our models for aggregate behavior, then we will be content to look for testable implications (and falsifications) of our models at the level of aggregate behavior.

The third line of defense is the most subtle. Even if we know that there are systematic violations of our models by individuals, violations that do not cancel out, we can still gain insight into questions of interest by studying models where we assume away those violations. This line of defense is delicate because it requires the theorist to have a deep understanding of which assumptions drive the conclusions that are generated by the theory; we will return to this in the next two sections.

The institutional framework

The actions taken by any individual depend on the opportunities that are presented to the individual. Those opportunities, in turn, often depend upon the collective actions of others. And the consequences for an individual of that individual's actions usually depend on what others have chosen to do. The term *institutional framework* is used throughout to refer to those parts of a model that describe (a) the general nature of options that an individual has, and (b) the options available to and outcomes ensuing for each individual, as a function of other individuals' actions.

In the traditional models of microeconomics, *prices in an impersonal marketplace* constitute the institutional framework; consumers can choose any bundle they can afford, where what is affordable is determined by prices. The market is impersonal in the sense that all consumers face the same array of prices. And the precise choices available to one consumer depend on the consumption choices of all consumers (and the production choices of firms) through these prices.

As we will discuss at length, the workings of the price mechanism are quite fuzzy and ambiguous. Where do prices come from? How are they set, and how do they reflect the actions of individual consumers? Note also the potential of circularity in this; prices constrain the choices of individual consumers, and those choices simultaneously determine prices.

Spurred by these questions, we will look at more concrete and detailed institutional frameworks that are more precise about the connection between the choices of some individuals and the options available to and outcomes resulting for other individuals. An example of a more concrete mechanism, which in some sense leads to the establishment of a "price," is the institution of a sealed bid, first price auction: Anyone who wishes to may submit a bid for an object, with the high bidder winning the object and paying his or her bid. A different institutional framework for the same purpose is a progressive oral auction — an auction where bids are made openly, and each new bid must beat the previous high bid by some minimal amount.

Predicting the outcome: Equilibrium analysis

Having modeled the behavior of individuals, the nature of the choices they have, and the ways in which their actions affect each other, we are left with the task of saying just what the product of these things will be, that is, predicting what actions will be selected and what results will ensue. We will use various forms of *equilibrium analysis* to address this task. Generally speaking, an equilibrium is a situation in which each individual agent is doing as well as it can for itself, given the array of actions taken by others and given the institutional framework that defines the options of individuals and links their actions.

Although it is not quite how we will conduct business, you might think of this as a feedback system; individuals make individual choices, and the institutional framework aggregates those actions into an aggregate outcome which then determines constraints that individuals face and outcomes they receive. If individuals take a "trial shot" at an action, after the aggregation is accomplished and the feedback is fed back, they may learn that their actions are incompatible or didn't have quite the consequences they foresaw. This leads individuals to change their individual actions, which changes the feedback, and so on. An equilibrium is a collection of individual choices whereby the feedback process would lead to no subsequent change in behavior.

You might suppose that instead of studying only the equilibria of such a system, we will dive in and look at the dynamics of the feedback, change in behavior, change in feedback, ..., cycle, that is, the *disequilibrium dynamics* for the institutions we pose. We will do something like this in a few cases. But when we do, we will turn it into a study of *equilibrium dynamics*, where the institutional framework is expanded to include the feedback loops that are entailed and where individuals' behaviors are dynamically in equilibrium — each individual doing as well as possible given the institutional framework and the dynamic behaviors of others.

Our use of equilibrium analysis raises the question: Why are equilibria germane? Put another way, why do we think that what happens corresponds to an equilibrium? *We make no such blanket assumption.* As we will make especially clear in chapter 12, we rely on intuition and experience to say when this sort of analysis is germane, and while we will sketch a variety of reasons that equilibria might be achieved and

equilibrium analysis might be relevant, we try never to *presume* in a particular application that this is so.

1.2. The purpose of microeconomic theory

Having set out the basic categories of microeconomic theory (at least as practiced in this book), we come to the question: What do we intend to get out of the theory? The simple answer is a better understanding of economic activity and outcomes.

Why do we seek a better understanding? One reason that needs no explanation is simple intellectual curiosity. But beyond that, a better understanding of economic activity can be useful in at least two ways. First, as a participant in the economic system, better understanding can lead to better outcomes for oneself. That is why budding business executives (are told to) study microeconomics; a better understanding of the ways in which markets work will allow them to make markets work better for themselves. And second, part of the study of microeconomics concerns the efficiency and specific inefficiencies in various institutional frameworks with a view towards policy. One tries to see whether a particular institution can, by tinkering or by drastic change, be made to yield a socially better outcome; the vague presumption is that changes that improve the social weal might be made via social and political processes. In this book, we will touch on the efficiency of various institutions, although this will be relatively deemphasized.

What constitutes better understanding? Put differently, how does one know when one has learned something from an exercise in microeconomic theory? The standard acid test is that the theory should be (a) testable and (b) tested empirically, either in the real world or in the lab.[1] But many of the models and theories given in this book have not been subjected to a rigorous empirical test, and some of them may never be. Yet, I maintain, models untested rigorously may still lead to better understanding, through a process that combines casual empiricism and intuition.

By casual empiricism joined with intuition I mean that the reader should look at any given model or idea and ask: Based on personal experience and intuition about how things are, does this make sense? Does it help put into perspective things that have been observed? Does it help organize thoughts about a number of "facts?" When and if so, the exercise of theory construction has been useful.

[1] See chapters 3, 6, and 15 if the concept of a laboratory test of an economic theory is new to you.

Imagine that you are trying to explain a particular phenomenon with one of two competing theories. Neither fits the data perfectly, but the first does a somewhat better job according to standard statistical measures. At the same time, the first theory is built on some hypotheses about behavior by individuals that are entirely ad hoc, whereas the second is based on a model of behavior that appeals to your intuition about how people act in this sort of situation. I assert that trying to decide which model does a better job of "explaining" is *not* simply a matter of looking at which fits better statistically. The second model should gain credence because of its greater face validity, which brings to bear, in an informal sense, other data.[2]

Needless to say, one's intuition is a personal thing. Judging models on the basis of their level of intuitive credibility is a risky business. One can be misled by models, especially when one is the creator of the model, and one is more prone to be misled the more complex the model is. Empirical verification should be more than an ideal to which one pays lip service; one should try honestly to construct (and even test) testable models. But intuition honestly applied is not worthless and should not be completely abandoned. Moreover, exercises can be performed to see what drives the conclusions of a model; examine how robust the results are to changes in specification and precise assumptions.

There is something of a "market test" here: one's ability to convince others of one's personal intuitive insights arising from specific models. Microeconomic theorists have a tendency to overresearch "fashionable" topics; insofar as they can be convinced by something because it is fashionable and not because it rings true, they are less than ideal for this market test. But less theoretically and more empirically inclined colleagues (and, sometimes even better, practitioners) are typically good and sceptical judges of the value of a particular model. Attempts to convince them, while sometimes frustrating, are usually helpful both to understand and improve upon a model.

The usefulness of falsified models

To push this even further, I would argue that sometimes a model whose predictions are clearly falsified is still useful. This can happen in at least three ways. First, insofar as one understands how the assumptions led to the falsified conclusions, one understands which assumptions *don't* lead to the "truth." Knowing what doesn't work is often a good place to begin to figure out what does.

[2] Readers who know about Bayesian inference should think of this as a case in which the likelihood of the first model is greater, but the second has higher prior probability. Which has a greater posterior probability is then unclear and depends on the relative strengths of the priors and likelihoods.

Second, theory building is a cumulative process. Under the presumption that most economic contexts share some fundamental characteristics, there is value in having models that conform to "generally accepted principles" that have served well in other contexts. Compared with a model that is radically different from standard models, a model that conforms to standard assumptions and principles will be better understood, both by the theory creator and by the audience. Moreover, unification of theory is valuable for its own sake to gain a better understanding of the shared characteristics. Of course, not all contexts are the same, and sometimes what economists think of as "generally acceptable and universally applicable principles" are pushed too far. Economists are well-known among social scientists as imperialists in the sense that economists attempt to reduce everything to economic notions and paradigms. But a real case still can be made for tradition and conservatism in the development of economic (and other) theory.

Granting this, when looking at an economic phenomenon that is poorly understood, economic theorists will attempt to build models that fit into the general rules of the discipline. Such attempts are rarely successful on the first trial. But insofar as first trials do lead to second and third trials, each of which gets one closer, the first trial is of value, and learning about another's first trial may help you construct the second and third.[3]

Economists tell a parable about theory and theorists that relates to this. An economic theorist lost his wallet in a field by a road and proceeded to search futilely for the wallet in the road under the illumination of a street light on the grounds that "that is where the light is." So, the folktale goes, it is with using theoretically "correct" but otherwise inappropriate theory. But an amendment to this tale is suggested by José Scheinkman: Without defending the actions of this particular theorist, perhaps one could try to construct a string of lights that begins in the road and that will eventually reach the field, on the grounds that the road is where electricity presently is found.

Third, and perhaps most subtly, models that fail to predict because they lack certain realistic features can still help clarify the analyst's thinking about the features they do encompass, as long as the analyst is able to combine intuitively and informally what has been omitted from the model

[3] That said, I cannot help but note that the journals contain many more papers that begin "This paper takes a first step in..." than papers that take second or third steps. Not every attempt at widening the scope of economic theory is successful!

with what has been learned from it. Of course, everything else held equal, it is "better" to have a model that captures formally more of the salient aspects of the situation than to have one that omits important features. But all else is rarely equal, and one shouldn't preclude building intuition with models that make somewhat falsified assumptions or give somewhat falsified conclusions, as long as one can understand and integrate informally what is missing formally.

1.3. Scope, detail, emphasis, and complexity

This leads to a third point about the theories and models that will be encountered in this book: their levels of scope, detail, emphasis, and complexity.

On scope, chapter 8 presents an excellent illustration. Chapter 8 concerns the theory of perfect competition. It begins with the analysis of a market for a single good — the theory of perfect competition that will be familiar to you from intermediate microeconomics. You may recall from intermediate micro that the industry supply curve is the horizontal sum of the individual supply curves of firms, and the supply curves of firms are their marginal cost curves. (If you don't recall this or never heard of it before, just read on for the flow of what I'm about to say.) This would lead to inaccurate predictions of, say, how price will move with shifts in demand, if one factor of production of the good in question comes from a market that itself has an upward sloping supply curve. To illustrate this, we move in chapter 8 from the traditional model of a single market to consideration of two linked markets, both of which are perfectly competitive. But why stop with two markets? Why not consider all markets together? At the end of chapter 8, under the assumption that all markets are perfectly competitive, we will see what can be said about such a (so-called) general equilibrium.

So what is the appropriate scope for a given model? There is a natural inclination to say that larger scope is better. But there are drawbacks to larger scope. The larger the scope, the more intractable the model, and the harder it is to draw out sharp and/or intuitively comprehensible results. One might feel relatively comfortable making strong assumptions in a model with a narrow scope about how the "outside" affects what is inside the model, assumptions that are (or seem) likely to be true and hence good modeling assumptions but that cannot easily be deduced generally in a model with an expanded scope.

For level of detail, consider how one should treat the family unit in a model of consumer demand. The standard model of consumer demand

envisions a single individual who maximizes his or her utility subject to a budget constraint. Almost anyone who has lived in a family will recognize that family expenditure decisions are a bit more involved than this. Individual members of the family, presumably, have their own tastes. But they recognize a strong interdependence in their personal consumption decisions. (Family members may like, or dislike, taking their vacations in the same location, for example.) And they often pool resources, so that their individual budget constraints get mixed together. Should we treat the family unit as a single consumer or attempt to build a more detailed model about interactions among family members? The answer will depend on what phenomena we are trying to explain. In most models, families are treated "coarsely," as consuming units, each family conforming to the standard model of the consumer. But for models of the labor supply decisions of families, for example, the detailed modeling of family interaction is sometimes crucial to what is being explained and so is included within the formal model. In general, one pays a cost in tractability and comprehension for increased detail. But in some contexts, greater detail is essential to get at the point in question.

Emphasis is meant to connote something like scope and like detail that isn't quite either. When looking at a particular phenomenon, the analyst may wish to emphasize certain aspects. To do this, the model may become unbalanced — heavy on one aspect of detail, or wide in one aspect of scope, but simple and narrow in all others. Especially when building a model to develop intuition concerning one facet of a larger problem, and where one will rely on less formal methods to integrate in other facets, models that (over)emphasize the facet of immediate interest are warranted.

This discussion is meant to alert the reader to a crucial aspect of microeconomic models and a crucial skill for microeconomists: Building a model is a matter of selecting the appropriate scope, level of detail, and matters to emphasize. There is no set formula to do this, because the appropriate levels of each depend on what the model seeks to accomplish. When "proving" the efficiency of the market system (see chapters 6 and 8), a wide scope is appropriate because overall efficiency is an economy-wide phenomenon. To demonstrate the *possibility* of a particular form of inefficiency in a particular form of market, narrower scope and greater emphasis on a few crucial details are called for. In this book you will see a wide variety of models that vary in their levels of scope, detail, and emphasis, and it is hoped that you will both get a taste for the variety of "types" of models that are used by microeconomic theorists and get a sense of the strengths and weaknesses of those types.

Let me add two further considerations, using the general notion of the level of complexity of one's models. Complexity combines scope and detail, but has other aspects as well, such as the level of mathematics required. Insofar as the model is meant to generate insights and to build intuition, too great a level of complexity in any form is usually bad. Insights and intuition are built from understanding — recognizing (in the model) which assumptions are crucial and what leads to what. The more complex a model is, the harder it is to achieve that sort of understanding. At the same time, simple models too often lead to flawed insights, because the conclusions depend too heavily on the simple setting that is assumed. Lessons that seem obvious from models of single markets in isolation are too easily overturned when one examines the interaction between markets. Balancing these considerations is not easy.

By the same token, building models to explain one's own insights to others requires a balance between complexity and simplicity. Recall that the proposed "market test" was whether you could convince an economist or practitioner who is perhaps not up-to-date with the latest fashion in economic theory but who instead has a well-developed intuitive understanding of how things are. Very complex models that require either a lot of knowledge about the latest techniques or incredible patience to work through mathematical detail will not pass this test. But, at the same time, a lot of the more complex techniques in microeconomic theory are there to check the *consistency* of models; however appealing the conclusions are, a model that is predicated on assumptions that are logically inconsistent or inconsistent with most standard theory is unsatisfactory. The best theories and models are those that pass (possibly complex) tests of consistency or validity and, at the same time, provide insights that are clear and intuitive and don't depend on some mathematical sleight-of-hand. Not all the theories and models that are presented here will pass this dual test. All will pass the consistency test (I hope), so you can anticipate that those that fail will fail on grounds of being too complex to be intuitive.[4] And those that do fail on grounds of complexity are, by virtue of that failure, somewhat of second (or lower) quality.

1.4. *A précis of the plot*

This book may be thought of as a collection of somewhat interconnected models that are interesting in their own right. But, at the same

[4] Put an X in the margin by this sentence. When we get to analysis of the centipede game in chapter 14, I will remind you that I gave fair warning.

time, the book has been written to convey the strengths and weaknesses of the tools that microeconomic theory has at its disposal. You should have little problem seeing the individual pieces from the table of contents, but it may be helpful to discuss the more global development that I intend.

First, insofar as one can split the subject of microeconomic theory into (1) actors and their behavior and (2) institutions and equilibrium, this is a book that stresses institutions and equilibrium. I don't mean to disparage the theories of individual behavior that are a cornerstone of microeconomic theory. I've even written a different book on one aspect of that subject. But I believe that *detailed* study of this topic can wait until one studies institutions and equilibrium, and the latter subject is of greater immediate interest and appeal. Still, it is necessary to understand the basics of the standard models of consumer (and firm) behavior before you can understand models of equilibrium, and part I and chapter 7 in part II provide the necessary background. Some readers of this book will have already taken a course that stressed the standard theories of the consumer and the firm. Such readers will probably find a light skim of chapters 2 and 7 adequate, but chapters 3, 4, and 5 present important background for the rest of the book that is treated here in greater depth than is typical in the traditional first course in microeconomics.

Part II takes up the classic "mechanism" by which individuals' diverse desires are meant to be brought into equilibrium: the price mechanism. We progress from situations of perfect competition in which, except for ambiguity about how prices are set, the price mechanism works both clearly and well, to monopoly and oligopoly, where it is less clear that prices alone are doing the equilibrating. By the time we reach the end of oligopoly, we see rather large holes in our understanding of how prices work, and we tentatively attribute those holes to a lack of specification of what precisely is "the price mechanism."

In part III we look at techniques that are meant to help us fill in those holes, techniques from noncooperative game theory. Game theoretic analyses have been extremely fashionable in microeconomic theory since the mid-1970s because such analyses are quite precise concerning institutional framework and because they help us see how institutions matter. Indeed, we will see cases in which, according to the theory, the institutional framework seems to matter too much. But we develop at the same time two weaknesses in such analyses. One, in many important cases, the theory is not determinative — many "equilibria" are possible, and which equilibrium, if any, arises depends on conjectures that are not provided for by the theory. Two, insofar as institutions matter, one has to wonder where existing institutions come from. Game theoretic analyses take the institutions as exogenously given. Until we understand where institutions come from and how they evolve, an important part of the puzzle is left unsolved.

Part IV exhibits some of the most important achievements of microeconomic theory in the 1970s and 1980s, developments in the so-called *economics of information*. You are meant to come away from part IV impressed by the insights that can be gleaned by the close analysis of market institutions, both with and without the techniques of noncooperative game theory. To be sure, the problems with our techniques that are discussed throughout part III are not resolved here, and many important questions are left unanswered. But insofar as microeconomic theory has made progress on substantial questions over the past twenty years, this is an area of progress.

In part V we take up the theory of the firm. In standard microeconomic theory firms are actors very much like consumers. We will criticize this classical view and consider instead a view of the firm as something in the category of a market — an institution within which diverse desires of consumers of various sorts are equilibrated. We are able to shed some light on what role firms and other nonmarket institutions play, but we will see that matters crucially important to this subject do not seem amenable to study with the tools at our disposal. In a sense we will come back full circle to individual behavior. The argument will be that we cannot do an adequate job studying firms and other institutions, and especially the origins and evolution of these institutions, unless and until we reformulate and refine the models we have of individual behavior.

All this is, presumably, a bit hard to fathom without a lot of fleshing out. But that's the point of the next eight hundred-odd pages.

part I

Individual and social choice

chapter two

The theory of consumer choice and demand

Prologue to part I

The central figure in microeconomic theory is the consumer, and the first step is almost always to provide a model of the consumer's *behavior*. This model may be substantially implied (see the discussion in 8.1 on demand curves), but it is there at some level. We usually think of the consumer as an entity who *chooses* from some given set of feasible options, and so our first order of business, which takes up most of part I, is to provide models of consumer choice.

You will recall from intermediate micro (and even most principles courses) the representation of consumer choice given in figure 2.1. We imagine a consumer who consumes two goods, say wine, denominated in numbers of bottles, and beer, denominated in numbers of cans. Imagine that this consumer has $1,000 to spend, the price of beer is $1 per can, and the price of wine is $10 per bottle. Then our consumer can purchase any combination of beer and wine in the shaded area in figure 2.1.

The consumer has preferences over combinations of beer and wine represented by *indifference curves*; these are the curves in figure 2.1. All

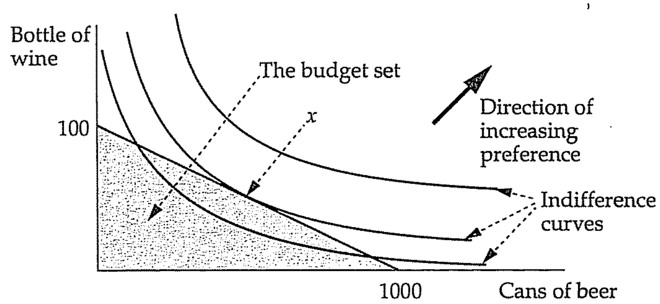

Figure 2.1. The usual picture for consumer demand.

points along a given curve are bundles of beer and wine that the consumer rates as being equally good (so she is *indifferent* among them), and the consumer prefers to be "further out," in the direction of the heavy arrow. Hence this consumer selects the point marked x, the point on the highest indifference curve that still is affordable given prices and the consumer's level of income.

With this picture, you and your teacher probably went on to consider how the consumer reacts as his level of income changes, the price of wine decreases, an excise tax is imposed on one good or the other, and so on. This picture encapsulates a simple model of consumer choice, viz., the consumer chooses from the feasible set some point that is on the highest available indifference curve. In this chapter, we explore the foundations for this picture (and more complex generalizations of it) by considering choice and preference in the abstract and we consider the fundamental application of this theory in microeconomics, *Marshallian demand*. In chapter 3, we consider choice where the objects from which choice is made have uncertain prizes. In chapter 4, we look at how to model choices that are made through time instead of all at once.

In most situations, when there is more than one consumer the choices the various consumers wish to make are somewhat in conflict. The rest of the book (chapter 7 excepted) concerns the resolution of such conflicts. We will, by and large, imagine that the conflicts are resolved by particular institutional arrangements (such as the price mechanism); we will go on to model specific institutions and study how the conflicts are resolved. But to close part I, we will look briefly at the problem of *social choice* in the abstract, studying principles that might be used to characterize particular resolutions and even to dictate a specific resolution.

To begin then, keep figure 2.1 firmly in mind as we ask, Where do these indifference curves come from and what do they represent?

2.1. Preferences and choices

The objects of choice

We are interested here in the behavior of an individual, called *the consumer*, who is faced with the problem of choosing from among a set of objects. To begin, we formalize the set from which the choice is to be made. Let X represent some set of *objects*. It is quite typical in economic applications to think of X as a space of consumption bundles, modeled as a subset of R^K, where R is the real line and K is the number of commodities. In an economy with three commodities, say beer, wine and whisky, $K = 3$, and a typical consumption bundle is a three-vector,

$x = (x_1, x_2, x_3)$, representing x_1 cans of beer, x_2 bottles of wine, and x_3 shots of whisky. The units — cans, bottles, shots — are arbitrary; we choose any convenient units.

A favorite game of economists is to make simple, basic constructions work in ever wider and wilder circumstances; while you think of commodities such as cans of beer, also think of commodities such as cans of beer next week, as distinct from cans of beer today or in two weeks time; the services of a bottle opener next week; the services of a bartender to draw a beer next week, if the temperature tomorrow gets over seventy degrees and then falls to sixty-five sometime in the next four days. These, and things even stranger, are shoehorned into the basic framework. For now, think of beer, wine, whisky (or, when we draw two-dimensional pictures as in figure 2.1, drop the whisky). But bear in mind that economic theory is often a shell game that seeks to convince you with simple, concrete examples and then gets you to extrapolate without thinking too much about more problematic settings.

The basic preference relation

The standard way to model the consumer is with a preference relation. Imagine that we present the consumer with pairs of alternatives, x and y, and ask how they compare. Our question is, Is either x or y better than the other in your eyes? If the consumer says that x is in fact better than y, we write $x \succ y$ and we say that x is *strictly preferred* to y.

For each pair x and y, we can imagine four possible responses to our question: (1) The consumer may say that x is better than y, but not the reverse. (2) She may say that y is better than x, but not the reverse. (3) She may say that neither seems better to her; she is unwilling to commit to a judgment. (4) She may say that x is better than y *and* y is better than x. Right away we wish to preclude the fourth possibility. It is logically possible for a consumer to state that each alternative is better than the other, but this stretches the meaning of the language too far and, more to the point, it would be very inconvenient for later purposes. Hence we assume

***Assumption 1.** Preferences are asymmetric: There is no pair x and y from X such that $x \succ y$ and $y \succ x$.*

But before you conclude that this is obvious, consider two problems with our entire story that bear on this assumption. First, we have said nothing about when our consumer is making these judgments. Our story sounds as if these judgments are made at some single point in time, in which case assumption 1 seems reasonable. But when we apply models that are based on assumption 1, we will sometimes assume that the judgments expressed by our consumer at one point in time remain valid at subsequent times. You might not find so reasonable an assumption that, if a consumer decides she will prefer x to y a week from now, four days hence she will necessarily continue to prefer x to y after three further days. But in some contexts, when consumers are making choices at a number of distinct dates, by making assumption 1 and applying the standard model as usual, we will be assuming that this is so. (For more on this see chapter 4.)

A second reason to wonder about assumption 1 concerns the framing of a particular choice. It is easiest to illustrate this with an example taken from Kahneman and Tversky (1979):

> As a doctor in a position of authority in the national government, you've been informed that a new flu epidemic will hit your country next winter and that this epidemic will result in the deaths of 600 people. (Either death or complete recovery is the outcome in each case.) There are two possible vaccination programs that you can undertake, and doing one precludes doing the other. The first will save 400 people with certainty. The second will save no one with probability 1/3 and 600 with probability 2/3. Which do you prefer?

Formulate an answer to this question, and then try:

> As a doctor in a position of authority in the national govenment, you've been informed that a new flu epidemic will hit your country next winter. To fight this epidemic, one of two possible vaccination programs is to be chosen, and undertaking one program precludes attempting the other. In the first program, 200 people will die with certainty. In the second, there is a 2/3 chance that no one will die, and a 1/3 chance that 600 will die. Which do you prefer?

These questions are complicated by the fact that they involve some uncertainty, the topic of chapter 3. But they make the point very well. Asked of medical professionals, the modal responses to this pair of questions were preferences for the first program in the first question and for the second program in the second question. The point is that the questions are identical in terms of outcomes — the only difference is in the way that the questions have been framed or phrased. In the first question, those 600 are dead, and most people choose to bring some of them back to life with

certainty. In the second question, no one is dead yet, and it seems rather callous to consign 200 people to certain death.[1] Framing can affect comparative judgments like the ones we are asking our consumer, especially when the items under consideration are complex. When items are complex, the cognitive process that underpins an eventual rank ordering can involve focusing on certain aspects, and the way in which the alternatives are framed can affect what is focused upon and, therefore, what is chosen. Given a particular framing of alternatives, we might not hear $x \succ y$ and $y \succ x$ in the same breath, but the pause between breaths in which these two "contradictory" judgments are made may be short indeed.

Despite these objections, assumption 1 is made in virtually every model of consumer choice in economics. Joined to it is a second assumption, which says that if a consumer makes the judgment $x \succ y$, she is able to place any other option z somewhere on the ordinal scale set by these two: better than y, or worse than x (or both). Formally:

Assumption 2. *Preferences are **negatively transitive**: If $x \succ y$, then for any third element z, either $x \succ z$, or $z \succ y$, or both.*

(The term *negative transitivity* will remain mysterious if you don't do problem 1.) The content of this assumption can best be illustrated by an example that is intended to cast doubt on it. Suppose that objects like x and y are bundles of cans of beer and bottles of wine: vectors (x_1, x_2), where the first component is the number of cans of beer and the second is the number of bottles of wine. Our consumer might have no problem comparing the two bundles $(20, 8)$ and $(21, 9)$. Presumably (assuming a lack of temperance), $(21, 9) \succ (20, 8)$. But let $x = (21, 9)$, $y = (20, 8)$, and $z = (40, 2)$ in assumption 2. Since $x \succ y$, the assumption maintains that either $x \succ z$, or $z \succ y$, or both; that is, either $(21, 9) \succ (40, 2)$, or $(40, 2) \succ (20, 8)$, or both. Our consumer, however, may claim that she can rank the two bundles $(21, 9)$ and $(20, 8)$, but comparing either with $(40, 2)$ is too hard, and our consumer refuses to commit to either ranking. It is this that assumption 2 rules out.

Beyond assumptions 1 and 2, other properties seem natural for strict preference. Three natural properties are:

(1) *Irreflexivity*: For no x is $x \succ x$.

(2) *Transitivity*: If $x \succ y$ and $y \succ z$, then $x \succ z$.

[1] If you care to see how even one word makes a difference, reread the first question with the word *all* inserted just before the 600 in the next-to-last sentence.

(3) *Acyclicity*: If, for a given finite integer n, $x_1 \succ x_2, x_2 \succ x_3, \ldots, x_{n-1} \succ x_n$, then $x_n \neq x_1$.

All three of these properties are implied by assumptions 1 and 2. We state this as a proposition:

✓ ***Proposition 2.1.*** *If $\succ$ is asymmetric and negatively transitive, then $\succ$ is irreflexive, transitive and acyclic.*

We leave almost all the proofs of propositions in this section to you, but this one is so easy that we will give it. First, irreflexivity is directly implied by asymmetry: If $x \succ x$, then $\succ$ would not be asymmetric. (Note that in the definition of asymmetry, we did not say that $x \neq y$.) For transitivity, suppose that $x \succ y$ and $y \succ z$. By negative transitivity and $x \succ y$, we know that either $x \succ z$ or $z \succ y$. But since $y \succ z$, asymmetry forbids $z \succ y$. Hence $x \succ z$. And, for acyclicity, if $x_1 \succ x_2$, $x_2 \succ x_3$, $\ldots$, $x_{n-1} \succ x_n$, then transitivity implies that $x_1 \succ x_n$. And then asymmetry implies that $x_1 \neq x_n$.

That is all there is to the basic standard model of preference in microeconomics: The consumer makes rank order judgments between pairs of alternatives that satisfy assumptions 1 and 2. From this point there are four paths to follow: First, from expressed strict preference $\succ$, economists create associated weak preference and indifference relations. Second, this model of rank ordering pairs of alternatives is related to choices made by the consumer, when she has more than two alternatives at her disposal. Third, for reasons of analytical convenience, we seek a numerical representation for preferences. And fourth, we seek additional assumptions that will make the model of individual preference (and resulting choice) more amenable to analysis, so, for example, we get pictures like that shown in figure 2.1, with "nicely" shaped indifference curves, where "nice" remains to be defined. We consider each of these in turn.

Weak preference and indifference

Suppose our consumer's preferences are given by the relation $\succ$. From this we can define two further relations among pairs of alternatives in X:

Definitions. *For x and y in X, write $x \succeq y$, which is read x is **weakly preferred** to y, if it is not the case that $y \succ x$. And write $x \sim y$, read as x is **indifferent** to y, if it is not the case that either $x \succ y$ or $y \succ x$.*

Note carefully what has been done here: Weak preference is defined as the absence of strict preference "the other way," and indifference is defined as

the absence of strict preference "in either direction." Does this accord with one's intuitive sense of what weak preference and indifference mean? Not entirely, insofar as there are judgments that are hard for the individual to make. Our consumer who is choosing cans of beer and bottles of wine may feel unable to say either $(40,2) \succ (20,8)$ or $(20,8) \succ (40,2)$, but this doesn't make her positively indifferent between these two. Or if it did, indifference would be a strange animal: If this implies $(20,8) \sim (40,2)$, and if a similar inability to rank $(21,9)$ and $(40,2)$ implies $(21,9) \sim (40,2)$, then two bundles ($(20,8)$ and $(21,9)$), one of which is strictly preferred to another, would both be indifferent to a third.

(*As promised in the preface, certain esoteric matters in this book will be relegated to paragraphs such as this one: tighter spacing, smaller type, and slight indentation. This material isn't required for most of what follows, and the reader may find it less confusing to skip these paragraphs on a first reading and then return to them later.*) In view of such concerns, we might be happier to define weak preference and indifference as follows: In addition to expressing strict preference $\succ$, the consumer expresses positive indifference $\sim$, and weak preference is defined as the conjunction of the two: $x \succeq y$ if either $x \succ y$ or $x \sim y$. We would probably want to insist that $x \sim y$ is incompatible with either $x \succ y$ or $y \succ x$, so by assumption 1, at most one of these three possibilities would be true for any pair x and y. But we do not preclude a fourth possibility, namely that x and y are simply incomparable. This is an interesting but nonstandard way to proceed, and we will not pursue it in depth here; see Fishburn (1970).

Despite such concerns, weak preference and indifference are usually defined from strict preference in the fashion of the definition given above. And if one has assumptions 1 and 2 in hand, the associated weak preference and indifference relationships are well behaved.

Proposition 2.2. *If strict preference $\succ$ is asymmetric and negatively transitive, and weak preference and indifference are defined from strict preference according to the definitions given above, then:*

*(a) Weak preference $\succeq$ is **complete**: For every pair x and y, either $x \succeq y$ or $y \succeq x$ or both.*

*(b) Weak preference $\succeq$ is **transitive**: If $x \succeq y$ and $y \succeq z$, then $x \succeq z$.*

*(c) Indifference $\sim$ is **reflexive**, $x \sim x$ for all x, **symmetric**, $x \sim y$ implies $y \sim x$, and **transitive**, $x \sim y$ and $y \sim z$ implies $x \sim z$.*[a]

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[a] (*And sometimes mathematical esoterica is found in footnotes. Footnotes that are labeled with letters are more technical. Footnotes labeled with numbers should be read by all.*) A binary relation that is reflexive, symmetric and transitive is called an *equivalence* relation in mathematical jargon. So we can paraphrase part (c) as: given our two assumptions concerning $\succ$, indifference is an equivalence relation. See problem 2 for more on this.

(d) If $w \sim x, x \succ y$, and $y \sim z$, then $w \succ y$ and $x \succ z$.

There is one thing left to say about strict and weak preference. We have taken strict preference as primitive (what the consumer expresses) and defined weak preference from it. Most books begin with weak preference and induce strict preference. I prefer beginning with strict preference because it makes it easier to discuss noncomparability possibilities. Most authors prefer weak preference, I suspect, because negative transitivity is hard to wrap one's head around. But in the standard treatment, these two approaches are completely equivalent. Suppose you started with weak preference $\succeq$ and then defined strict preference $\succ$ by $x \succ y$ if it is not the case that $y \succeq x$ and defined indifference by $x \sim y$ if both $x \succeq y$ and $y \succeq x$. Then there would be no difference between beginning with $\succ$ or with $\succeq$. Moreover, if you began with $\succeq$, which is complete and transitive, the relation $\succ$ obtained would be asymmetric and negatively transitive. So for the standard treatment, it makes no difference whether you begin with asymmetric and negatively transitive strict preference $\succ$ or with complete and transitive weak preference $\succeq$, as long as you define the other as above.[b]

When we discussed negative transitivity of $\succ$, we illustrated what could go wrong with an example suggested by an inability to compare bundles. If we had begun our discussion of preferences with $\succeq$, it would have been natural to use a similar example to show what goes wrong with an assumption that $\succeq$ is complete. You may conclude from this that negative transitivity of $\succ$ corresponds in some way to completeness of $\succeq$. Yet if you do problem 1, you will see that this is wrong; negative transitivity of $\succ$ corresponds to transitivity of $\succeq$, while asymmetry of $\succ$ corresponds to completeness of $\succeq$. The reason for this apparent conundrum is that we define $x \succeq y$ if it is not the case that $y \succ x$ (and vice versa). In this definition, when we start with $\succ$, we include in $\succeq$ any cases of noncomparability. Put differently, if we admitted the possibility of incomparable pairs and so defined separately from $\succ$ a relation $\sim$ that expresses positive indifference, and we then defined $\succeq$ not by $x \succeq y$ if not $y \succ x$, but instead by $x \succeq y$ if $x \succ y$ or $x \sim y$, the relationship between negative transitivity of $\succ$ and completeness of $\succeq$ would be more in line with our intuition. (You will probably make sense of all this only if you take the time to hunt down a treatment of preference in which there is a formal difference between positive indifference and noncomparability.)

From preference to choice

So far our consumer has given her rank ordering between pairs of alternatives out of X. In typical situations, she will be choosing out of a

[b] When authors begin with $\succeq$, they sometimes add a third assumption, namely that $\succeq$ is reflexive, or $x \succeq x$ for all x. This is implied by completeness and so is redundant.

set with more than two members. So how do we relate her preferences and her choice behavior? It is typical to assume that *choice* is induced from *preference*, according to the following formal definition.

Definition. *Given a preference relation $\succ$ on a set of objects X and a nonempty subset A of X, the* ***set of acceptable alternatives*** *from A according to $\succ$ is defined to be*

$$c(A;\succ) = \{x \in A : \text{there is no } y \in A \text{ such that } y \succ x\}.$$

The content of this definition is that the consumer is happy to choose anything that isn't bettered by something else that is available. Note several things concerning this:

(1) The set $c(A;\succ)$ is, by definition, a subset of A. Anything else wouldn't make sense given our interpretation.

(2) The set $c(A;\succ)$ may contain more than one element. When $c(A;\succ)$ contains more than one element, the interpretation is that the consumer is willing to take any one of those elements; she isn't particular about which one she gets.[2]

(3) In some cases, the set $c(A;\succ)$ may contain no elements at all. For example, suppose that $X = [0,\infty)$ with $x \in X$ representing x dollars. And suppose A is the subset $\{1,2,3,\ldots\}$. If you always prefer more money to less, or $x \succ y$ whenever $x > y$, $c(A;\succ)$ will be empty. No matter how much money you might take, some amount of money in A is greater, hence better. Or suppose, in the same context, that $A = [0,10)$. For those of you unused to this notation, this means that A consists of all dollar amounts up to *but not including* 10. Then if money is divisible into arbitrarily fine units, there is no element of A that isn't bettered by some other.[c]

[2] Students will sometimes get a bit confused at this point, so let me be very pedantic. In most applications, an alternative x is a bundle of goods — a vector representing so many bottles of wine, so many cans of tuna fish, etc. If $c(A;\succ)$ has two elements, say x and x', we do not interpret this as saying that the individual chooses to take both x and x' together. If having both these bundles together was a possibility, then the vector sum $x + x'$ would be an element of A. Instead, $c(A;\succ) = \{x, x'\}$ means that the individual is happy with *either x or x'*.

[c] You may object that (a) there isn't an infinite amount of money to be had and (b) money isn't divisible past the penny or the mill, so that both examples are unrealistic. In this case, let me pose one further example. I will flip a coin N times, where N is a positive integer. On the first occurrence of heads, I will give you \$10. If there are N tails in a row, you get nothing. You must choose N from the set $\{1,2,\ldots\}$. Now you might claim that you are indifferent between $N = 10^{100}$ and anything larger. But if you always prefer N to be larger, then you will have no best choice.

(4) In the examples just given, $c(A;\succ)$ is empty because A is too large or "not nice." In the next section, we will see how to avoid this problem (by restricting the range of sets A to which $c(\cdot;\succ)$ is applied). But we could also get empty choice sets if $\succ$ is badly behaved; suppose $X = \{x, y, z, w\}$, and $x \succ y$, $y \succ z$, and $z \succ x$. Then $c(\{x,y,z\};\succ) = \emptyset$.

(5) On an entirely different plane of objections to this approach, you might expect that what our consumer chooses out of a set of alternatives A depends on such things as the time of day, the way the choices are framed, or even the range of choices offered; she chooses apple pie over cherry, unless she can alternatively have banana, in which case cherry is chosen. All these sorts of choice behaviors are observed in real life, and they are *not* modelable given our preference-based definition of choice. We will return to this.

We haven't yet said anything about the consequences for choice of assuming that $\succ$ conforms to the standard assumptions. We can add that to the stew at this point:

Proposition 2.3. *Suppose that $\succ$ is asymmetric and negatively transitive. Then*

(a) For every finite set A, $c(A;\succ)$ is nonempty.

(b) Suppose that both x and y are in both A and B, and $x \in c(A;\succ)$ and $y \in c(B;\succ)$. Then $x \in c(B;\succ)$ and $y \in c(A;\succ)$.

You are asked to provide a proof in problem 3. This proposition leaves open the question about choice from infinite sets A. We will deal with this to some extent below.

From choice to preference

A quick review of the plot so far is: We started with strict preferences and saw connections with weak preference and indifference. Then we discussed how choice behavior might be induced from preferences. But from the viewpoint of descriptive theory (where one tries to build a model that matches what is observed), one would naturally proceed the other way around. Unless we provide our consumer with a questionnaire or otherwise directly enquire as to her preferences, the only signs we will see of her preferences are the choices she makes. Put another way, saying that the consumer has preferences that explain choice according to the formula given in the previous subsection is our *model* of her behavior, and it would make more sense to take her behavior as primitive and ask, When are her choices consistent with our preference-based model of choice?

In an abstract and not entirely satisfactory sense, we take up this question next. The approach is not entirely satisfactory because of the primitive we use; we assume that we have our hands on the consumer's entire *choice function.* Consider the following formal definition.

Definition. *A **choice function** on X is a function c whose domain is the set of all nonempty subsets of X, whose range is the set of all subsets of X, and that satisfies $c(A) \subseteq A$ for all $A \subseteq X$.*

To repeat, the interpretation is that the consumer, given her choice out of A, is content to take any one element of $c(A)$. Compare this definition with the definition of $c(\cdot,\succ)$ that was given for a preference relation $\succ$. Viewed as a mathematical object, $c(\cdot,\succ)$ is a choice function for fixed $\succ$. That is, in the previous subsection we constructed a choice function (according to the definition just given) from a primitive preference relation. In this subsection, the choice function is the primitive object.

We can raise objections to this approach similar to those raised concerning $\succ$; taking choice as a primitive in this fashion doesn't pay heed to the possibility that choice will change with changes in the timing of choice or the framing of alternatives. As before, economists for the most part ignore these possibilities. And they go on, usually, to make the following two assumptions about c (which you should be comparing with the two parts of proposition 2.3).

Assumption 3. *The choice function c is nonempty valued: $c(A) \neq \emptyset$ for all A.*

Assumption 4. *The choice function c satisfies **Houthakker's axiom of revealed preference:** If x and y are both contained in both A and B and if $x \in c(A)$ and $y \in c(B)$, then $x \in c(B)$ and $y \in c(A)$.*

Assumption 3 has already been discussed in the context of $c(\cdot,\succ)$ for a given $\succ$, but let us say a bit more about it. The problems encountered when a set A is infinite remain, and so we restrict attention for the time being to the case of finite X. Even granting this assumption, when choice is primitive, we can once again wonder about our consumer's actions when faced with a difficult decision. Consider our consumer choosing cans of beer and bottles of wine, and suppose she is given her choice out of $\{(40,2),(20,9)\}$. If she has difficulty comparing these two bundles, she might freeze up and be unable to choose. This is precluded by assumption 3. We don't rule out the possibility that she chooses "both"; that is, $c(\{(40,2),(20,9)\}) = \{(40,2),(20,9)\}$, which means that she would be happy with either of the two bundles. But we would interpret this as a definite expression of indifference between the two bundles.

Next examine assumption 4. It bears the following interpretation. Under the conditions that x and y are both in A and $x \in c(A)$, our consumer *reveals* by this that x is (weakly) preferred to y. Now if $y \in c(B)$ and $x \in B$, since x is no worse than y, x should also be among the most preferred things in B. (And by a symmetric argument, $y \in c(A)$ must hold as well.) Once again, the easiest way to grasp what this is saying is to consider a case in which it would fail. Suppose that x represents a wonderful meal at a classical French restaurant, while y represents an equally wonderful meal at a classical Japanese restaurant. The choice is a difficult one, but our consumer, given the choice out of $A = \{x, y\}$, decides that she prefers x. That is, $c(\{x, y\}) = \{x\}$. Now imagine that z is a meal at a somewhat lesser French restaurant, one at which our consumer had the misfortune recently to dine and which caused her an upset stomach from all the rich food. Suppose that $B = \{x, y, z\}$. Now if we give our consumer her choice out of B, the presence of z among the alternatives may remind her of all the butter and cream that goes into classical French cooking and that didn't agree with her when she dined at z. This, in turn, colors her evaluation of x; perhaps she prefers Japanese cuisine after all. This might lead to behavior given by $c(B) = \{y\}$, which together with $c(A) = \{x\}$ constitutes a violation of assumption 4. This is something like a framing effect; the presence of other alternatives may change how one regards a given alternative and thus change one's choices. It is this that assumption 4 rules out.

An interesting interplay existing between properties of nonemptiness and revealed preference is worth mentioning. By assuming nonemptiness, we rule out the possibility that the consumer, faced with a set A, is unable to choose. We could reduce this to a tautology by saying that choosing not to choose is a choice; that is, append to the set A one more element, "make no choice," and then whenever choice is too hard to make, interpret it as a choice of this "make no choice" element. If we did proceed by allowing "make no choice" to be a possible item, we would run into two problems. First, c is supposed to be defined on all subsets of X, and we've just assumed that it is defined only on subsets that contain this "make no choice" option. But we can deal with c defined only on some subsets of X; cf. problem 4(c). Second, and more substantively, if nonemptiness is really a problem, we are probably going to run afoul of the axiom of revealed preference by this dodge. Suppose that x and y are so hard to rank that the consumer freezes when given a choice out of $\{x, y\}$. We would, in our tautologizing, interpret this as saying that $c(\{x, y, \text{"make no choice"}\}) = \{\text{"make no choice"}\}$. But one assumes that $c(\{x, \text{"make no choice"}\}) = \{x\}$; if x is the only thing really on offer, it is taken, at least if it is not noxious. This, then, would constitute a violation of the axiom of revealed preference.

On the same point, imagine an indecisive consumer who wishes to obey assumption 3. This consumer, when faced with a difficult choice, gets past her indecision by volunteering to take any option that she can't disqualify. Our wine-and-beer-drinking friend will illustrate the point. She has a hard time comparing the two bundles (40, 2) and (20, 8), so her choice function specifies $c(\{(40,2), (20,8)\}) = \{(40,2), (20,8)\}$. And she has a hard time comparing the two bundles (40, 2) and (21, 9), so her choice function specifies $c(\{(40,2), (21,9)\}) = \{(40,2), (21,9)\}$. But then imagine giving her a choice out of the threesome $\{(40,2), (21,9), (20,8)\}$. Presumably she chooses either of the first two of these three — the bundle (20, 8) is excluded because (21, 9) is surely better. But if this is her choice function evaluated on the threesome, she has run afoul of Houthakker's axiom. (Prove this!) Salvation for an indecisive consumer is not possible according to the standard model.

We have already seen in proposition 2.3 that if our consumer has strict preferences given by an asymmetric and negatively transitive relation $\succ$, and if we define a choice function $c(\cdot;\succ)$ from $\succ$ (on a finite set X), then this choice function satisfies assumptions 3 and 4. The converse is also true, in the following sense: Given a choice function c, define an induced strict preference relation $\succ_c$ by $x \succ_c y$ if for any $A \subseteq X$ with $x, y \in X$, $x \in c(A)$ and $y \notin c(A)$. In words, $x \succ_c y$ if in any instance when both x and y are available, x is chosen and y isn't. Or, put succinctly, $x \succ_c y$ if choice ever reveals an instance where x seems to be strictly preferred to y.

Proposition 2.4. *Given a choice function c that satisfies assumptions 3 and 4, the relation $\succ_c$ defined from c as above is asymmetric and negatively transitive. Moreover, if you begin with c satisfying assumptions 3 and 4, induce $\succ_c$, and then define $c(\cdot;\succ_c)$, you will be back where you started: $c(\cdot;\succ_c) \equiv c$. And if you begin with $\succ$, induce $c(\cdot;\succ)$, and then induce from this $\succ_{c(\cdot;\succ)}$, you will again be back where you started.*

That is, taking as primitive $\succ$ that is asymmetric and negatively transitive is equivalent to taking as primitive a choice function c that is nonempty valued and satisfies Houthakker's axiom of revealed preference. The proof is left as an exercise.

There are two remarks to make here. First, proposition 2.4 is true under the maintained hypothesis that X is finite. Extensions to infinite X can be developed, but then one has to be careful about nonemptiness of the choice function. Except for brief remarks in the next section, we won't take up this subject here, leaving it for the interested reader to explore in more advanced treatments of the subject. (Look for references to abstract revealed preference theory.) Second, proposition 2.3 can be viewed as providing the testable restriction of the standard model of choice arising from preference; one needs to look either for a violation of nonemptiness or of Houthakker's axiom of revealed preference. As long as the data are consistent with a choice function that satisfies these two assumptions, they are consistent with the standard preference-based theory of consumer behavior. (But see problem 4 for more on this point.)

We will return to revealed preference interpreted as *the* testable restriction of the standard model of consumer behavior in the next section, when we get to the specific application of consumer demand.

Utility representations

Our third excursion from the starting point of consumer preferences concerns *numerical representations*. We need a definition.

Definition. *Given preferences $\succ$ on a set X, a* **numerical representation** *for those preferences is any function U with domain X and range the real line such that*

$$x \succ y \quad \textit{if and only if} \quad U(x) > U(y).$$

That is, U measures all the objects of choice on a numerical scale, and a higher measure on the scale means the consumer likes the object more. It is typical to refer to such a function U as a *utility function* for the consumer (or for her preferences).

Why would we want to know whether $\succ$ has a numerical representation? Essentially, it is convenient in applications to work with utility functions. As we will see in later sections of this chapter and in the problems, it is relatively easy to specify a consumer's preferences by writing down a utility function. And then we can turn a choice problem into a numerical maximization problem. That is, if $\succ$ has numerical representation U, then the "best" alternatives out of a set $A \subseteq X$ according to $\succ$ are precisely those elements of A that have maximum utility. If we are lucky enough to know that the utility function U and the set A from which choice is made are "nicely behaved" (e.g., U is differentiable and A is a convex, compact set), then we can think of applying the techniques of optimization theory to solve this choice problem.

All sorts of nice things might flow from a numerical representation of preference. So we want to know: For a given set of preferences $\succ$, when do these preferences admit a numerical representation?

Proposition 2.5. *For $\succ$ to admit a numerical representation, it is necessary that $\succ$ is asymmetric and negatively transitive.*

But these two properties are not quite sufficient; you need as well that either the set X is "small" or that preferences are well behaved.[d] We will give the two simplest and most useful "converses" to proposition 2.5, although the second requires that you know a bit of mathematics and so is treated as optional reading. You should have no problem proving both propositions 2.5 and 2.6. Consult a good book on choice and utility theory for a proof of 2.7 and for variants on 2.7.

Proposition 2.6. *If the set X on which $\succ$ is defined is finite, then $\succ$ admits a numerical representation if and only if it is asymmetric and negatively transitive.*[e]

As for the second converse to proposition 2.5, we will specialize to the case where the objects being chosen are consumption bundles: elements of R^K, for some K. We assume that negative consumption levels are not possible, so X is the positive orthant in R^K.[3] In this setting, we say that preferences given by $\succ$ are *continuous*

(a) if $\{x^n\}$ is a sequence of consumption bundles with limit x, and if $x \succ y$, then, for all n sufficiently large, $x^n \succ y$

(b) and if $\{x^n\}$ is a sequence of consumption bundles with limit x, and if $y \succ x$, then, for all n sufficiently large, $y \succ x^n$

Proposition 2.7. *In this setting, if $\succ$ is asymmetric, negatively transitive, and continuous, then $\succ$ can be represented by a continuous function U. Moreover, if $\succ$ is represented by a continuous function U, then $\succ$ must be continuous as well as asymmetric and negatively transitive.*

The restriction to X being the positive orthant in R^K for some K isn't really important to this result. We do need some of the topological properties that this setting provides, but the result generalizes substantially. See Fishburn (1970). Or if you prefer reading original sources, see Debreu (1954).

Suppose that $\succ$ has a numerical representation U. What can we say about other possible numerical representations? Suppose that $f : R \to R$ is a strictly increasing function (some examples: $f(r) = r^3 + 3r^5$; $f(r) = e^r$; $f(r) = -e^{-r^3}$). Then defining a function $V : X \to R$ by $V(x) = f(U(x))$ gives another numerical representation for $\succ$. This is easy to see: $V(x) > V(y)$ if and only if $U(x) > U(y)$. Since V and U induce the same order on X, whatever U represents (ordinally) is represented just as well by V.

[d] See problem 5 for an example of what can go wrong.

[e] In fact, this is true as long as X is countably infinite.

[3] We will try to be careful and distinguish between numbers that are *strictly positive* and those that are *nonnegative*. But much of conventional language blurs these distinctions; in particular, the *positive* orthant of R^K is used when referring to vectors that can have zeros as components. We will signal the other cases in which following conventional language leads to this sort of ambiguity.

The converse to this isn't true: That is, if V and U both represent $\succ$, there is not necessarily a strictly increasing function $f : R \to R$ with $V(x) = f(U(x))$ for all x. But almost this is true. There is always a function $f : R \to R$ that is nondecreasing and strictly increasing on the set $\{r \in R : r = U(x) \text{ for some } x \in X\}$ such that $V(\cdot) \equiv f(U(\cdot))$.[f] A rough paraphrase of the situation is

Numerical representations for $\succ$ are unique only up to strictly increasing rescalings.

The point is that the units in a utility scale, or even the size of relative differences, have no particular meaning. We can't, in looking at a change from x to y, say that the consumer is better off by the amount $U(y) - U(x)$ or by anything like this. At this point (it will be different when we get to uncertainty in chapter 3), the utility function is introduced as an analytical convenience. It has no particular cardinal significance. In particular, the "level of utility" is unobservable, and anything that requires us to know the "level of utility" will be untestable. This is important as we go through demand theory; we'll want to be careful to note which of the many constructions we make are based on observables, and which are things that exist (if at all) only in the mind of the economist.

Properties of preferences for consumption bundles

When (momentarily) we go off into the subject of consumer demand, we will be making all manner of assumptions about the utility function that represents our consumer's preferences. We might ask, for example, exactly what it takes to get pictures as "nice" as in figure 2.1.

Since our interest attaches specifically to the problem of consumer demand, we will specialize throughout this subsection to the case that X is the positive orthant of R^K for some integer K. That is, there are K commodities, and preferences are defined over bundles of those commodities where the amount of each commodity in any bundle is required to be nonnegative.[g] We assume throughout that preferences $\succ$ for some consumer are given, and these preferences are asymmetric and negatively transitive.

We can, with this alone, begin to offer some interpretation of figure 2.1. For each $x \in X$, we define the *indifference class* of x, denoted

[f] If you are good at mathematics, proving this should be easy.

[g] For those readers who know some real analysis, extensions to linear spaces for X should not prove difficult.

Indiff(x), by Indiff(x) $= \{y \in X : y \sim x\}$. Since $\sim$ is reflexive, symmetric, and transitive, we can show that the family of indifference classes, or the various Indiff(x) ranging over x, partition X. That is, every $y \in X$ is in one and only one Indiff(x). You are asked to prove this in problem 2. The indifference curves of figure 2.1 are then indifference classes for some implicit preference ordering of the consumer under investigation.

The questions we wish to address in this context are: What further properties might we think reasonable to suppose of $\succ$? And how do those properties translate into properties of numerical representations for $\succ$ and into pictures such as figure 2.1?

Monotonicity and local insatiability. In many cases, it is reasonable to assume that consumers prefer more to less. Or at least they do not strictly prefer less to more. We have the following definitions and results:

Definitions. *Preferences $\succ$ are* **monotone** *if for any two bundles x and y such that $x \geq y$, $x \succeq y$. (By $x \geq y$, we mean that each component of x is at least as large as the corresponding component of y.) And preferences $\succ$ are* **strictly monotone** *if for any two bundles x and y such that $x \geq y$ and $x \neq y$, $x \succ y$.*

A function $U : X \to R$ is **nondecreasing** *if for any two bundles x and y such that $x \geq y$, $U(x) \geq U(y)$. And U is* **strictly increasing** *if for any two bundles x and y such that $x \geq y$ and $x \neq y$, $U(x) > U(y)$.*

Proposition 2.8. *If U represents preferences $\succ$, these preferences are monotone if and only if U is nondecreasing, and these preferences are strictly monotone if and only if U is strictly increasing.*

The proof of this proposition is virtually a matter of comparing definitions.

One might wish somewhat less of some goods; week-old fish and contaminated water come to mind. For others some of the good is desirable, but past a certain point any additional amounts are undesirable; any extremely rich or sweet food would be a natural example. One doesn't value more of many goods; while I would not mind having around more hydrochloric acid or rolled steel, I do not strictly prefer more of these goods to less. The first two sorts of examples cast doubt on an assumption that preferences are monotone, while the last example questions strict monotonicity. Neither assumption is necessary for much of what we do subsequently. But for later purposes we need the assumption that arbitrarily close to any bundle x is another that is strictly preferred to x. This property is called *local insatiability*.

Definition. *Preferences $\succ$ are* ***locally insatiable*** *if for every $x = (x_1, \ldots, x_K) \in X$ and for every number $\epsilon > 0$ there is another bundle $y = (y_1, \ldots, y_K) \in X$ such that (a) $|x_j - y_j| < \epsilon$ for every $j = 1, \ldots, K$, and (b) $y \succ x$.*[h]

The interested reader can translate this property into the corresponding statement about a numerical representation of $\succ$.

Convexity. The next property that we consider is *convexity* of preferences.

Definitions. *(a) Preferences $\succ$ are* ***convex*** *if for every pair x and y from X with $x \succeq y$ and for every number $a \in [0,1]$, the bundle $ax + (1-a)y \succeq y$.*

(b) Preferences $\succ$ are ***strictly convex*** *if for every such x and y, $x \neq y$, and for every $a \in (0,1)$, $ax + (1-a)y \succ y$.*

(c) Preferences $\succ$ are ***semi-strictly convex*** *if for every pair x and y with $x \succ y$ and for every $a \in (0,1)$, $ax + (1-a)y \succ y$.*

By $ax + (1-a)y$, we mean the component-by-component convex combination of the two bundles.

Why would one ever think that preferences are or should be convex? The story, such as it is, is related to the notion of diminishing marginal utility or the classic ideal of "moderation in all things." Under the assumption $x \succeq y$, we know that in moving along the line segment from y to x we will reach a point (x) at least as good as the point (y) from which we started. The various forms of convexity are variations on the general notion that at each step along this path, we are never worse off than where we began. That, precisely, is convexity. Semi-strict convexity maintains that if $x \succ y$, so we will be better off at the end of the journey, then we are strictly better off at each step. And strict convexity holds that even if $x \sim y$, if we are strictly between the two we are better off than at the extremes.

You will sometimes see convexity of preferences defined a bit differently. For each point $x \in X$, define the set $\text{AsGood}(x) = \{y \in X : y \succeq x\}$. Recall that a set $Z \subseteq R^k$ is *convex* if for every x and y from Z and every number $a \in [0,1]$, $ax + (1-a)y \in Z$. (If you have never seen or heard of

[h] Note that becoming satiated in one commodity, say baklava, does not necessarily pose problems for local insatiability; all that is needed is that, from any consumption bundle, the consumer would prefer a small increase (or decrease) in *some* of the commodities.

the definition of a convex set before, get someone who has seen it to draw a few pictures for you.) Then:

Proposition 2.9. *Preferences $\succ$ are convex if and only if for every point x, the set AsGood(x) is convex.*

The proof of this proposition is left as an exercise.

As for the consequences of convexity of preferences for numerical representations, we must give some definitions from mathematics. These definitions are used throughout this book, so pay attention if they are new to you:

Definitions. *Let Z be a convex subset of R^K for some integer K and let f be a function with domain Z and range R.*

The function f is ***concave*** *if for all $x, y \in Z$ and $a \in [0,1]$, $f(ax+(1-a)y) \geq af(x) + (1-a)f(y)$. This function is* ***strictly concave*** *if for all such x and y, $x \neq y$, and for all $a \in (0,1)$, $f(ax+(1-a)y) > af(x) + (1-a)f(y)$.*

The function f is ***convex*** *if for all $x, y \in Z$ and $a \in [0,1]$, $f(ax+(1-a)y) \leq af(x) + (1-a)f(y)$. This function is* ***strictly convex*** *if for all such x and y, $x \neq y$, and for all $a \in (0,1)$, $f(ax+(1-a)y) < af(x) + (1-a)f(y)$.*

The function f is ***quasi-concave*** *if for all $x, y \in Z$ such that $f(x) \geq f(y)$ and for all $a \in [0,1]$, $f(ax+(1-a)y) \geq f(y)$. This function is* ***strictly quasi-concave*** *if for such x and y, $x \neq y$, and for $a \in (0,1)$, $f(ax+(1-a)y) > f(y)$. And f is* ***semi-strictly quasi-concave*** *if for all x and y with $f(x) > f(y)$ and for all $a \in (0,1)$, $f(ax+(1-a)y) > f(y)$.*

The function f is ***quasi-convex*** *if for all $x, y \in Z$ such that $f(x) \geq f(y)$ and for all $a \in [0,1]$, $f(ax+(1-a)y) \leq f(x)$. This function is* ***strictly quasi-convex*** *if for such x and y, $x \neq y$, and for $a \in (0,1)$, $f(ax+(1-a)y) < f(x)$.*[i]

Proposition 2.10. *(a) If preferences $\succ$ are represented by a concave function U, then preferences are convex. If they are represented by a strictly concave function U, then they are strictly convex.*

(b) Suppose that U is a numerical representation of preferences $\succ$. Then U is quasi-concave if and only if preferences $\succ$ are convex; U is strictly quasi-

[i] We have no need of semi-strictly quasi-convex functions. We also didn't define semi-strict concavity for functions. Try to draw a concave function that isn't semi-strictly quasi-concave and you'll see why.

concave if and only if preferences $\succ$ are strictly convex; and U is semi-strictly quasi-concave if and only if preferences $\succ$ are semi-strictly convex.[4]

Once again, the proofs are left to you. Note that part (a) runs in one direction only; if preferences have a concave representation, they are convex. But convex preferences can have numerical representations that are not concave. We can demonstrate this quite simply: Suppose that U is a concave function that represents $\succ$. We know from the discussion following propositions 2.5 and 2.6 that if $f : R \to R$ is a strictly increasing function, then the function V defined by $V(x) = f(U(x))$ is another numerical representation of $\succ$. But it is quite easy to construct, for a given concave function U, a strictly increasing function f such that $f(U(\cdot))$ is not concave. Create such an example if this isn't obvious to you. The idea is to make f "more convex" than U is concave.

In contrast, part (b) says that every representation of convex preferences $\succ$ is quasi-concave. Hence we conclude that if U is a quasi-concave function and f is strictly increasing, $f(U(\cdot))$ is also quasi-concave. (It is equally clear that similar statements hold for strict and semi-strict quasi-concavity.) You may find it helpful to prove this directly, although you should be sure that you understand the indirect path of logic that got us to this conclusion.

This leaves open one question. We know (or, rather, you know if you constructed an example two paragraphs ago) that convex preferences can have numerical representations that are not concave functions. But this doesn't mean that we can't show: *If preferences $\succ$ are convex, they admit at least one concave numerical representation.* But, in fact, we can't show this; it is quite false. On this point, see problem 7.[j]

Continuity. We have already mentioned the property of continuous preferences. It was used previously as a way to guarantee that preferences have a numerical representation when the set X is not finite, and in fact we claimed that it guaranteed the existence of a *continuous* numerical representation. We don't need continuity of preferences to guarantee the existence of a numerical representation. But, at the same time, continuity has intuitive appeal in its own right. N.B., continuous preferences always admit a continuous numerical representation. But not every numerical representation of continuous preferences is continuous.

[4] You are entitled to a bit of displeasure concerning the fact that *convex* preferences go together with (quasi-)*concave* utility functions. This would all be easier to remember if we called the property of preferences concavity. But, as you will see in chapter 6, convexity of the AsGood sets plays the crucial mathematical role, and for this reason preferences with convex AsGood sets are said to be convex, even though they have (quasi-)concave representations.

[j] In chapter 3, we will be able to give an intuitive property about the consumer's preferences that ensures these preferences admit a concave numerical representation.

About figure 2.1. The preferences that are implicitly identified by the very standard picture in figure 2.1 are strictly monotone and strictly convex. You can see the strict convexity from the fact that the indifference curves are strictly convex to the origin; if we take any two points and the line segment between them, this segment will be everywhere above the indifference curve of the "lower" of the two points. Strict monotonicity is implicit from the fact that the curves run down and to the right, never flattening out to be completely horizontal or vertical, so that from any point, any second point above and/or to the right is on a strictly "higher" indifference curve.

And the preferences represented in figure 2.1 are continuous. We see this because the "weakly better than" and "weakly worse than" sets for each point x are all closed; you can check that this is equivalent to the definition of continuity of preferences that we gave.

The technically astute reader may wish to investigate the extent to which figure 2.1 is *the* picture for preferences that are continuous, strictly convex and strictly monotone. The reader may also wish to investigate how the picture might change if we drop strict convexity or even convexity altogether, or if we drop (strict) monotonicity. On this general question, see the alternative pictures appearing later in this chapter.

2.2. Marshallian demand without derivatives

The prototypical application of the general model of choice that we have developed is *Marshallian demand*. The theory of Marshallian (consumer) demand has played a central role in the development of microeconomic theory; it is traditionally viewed as among the most important subjects to study in microeconomics, and the results obtained are useful in many applied contexts.

A complete and precise treatment of Marshallian demand takes many pages and much time. Because Marshallian demand plays a very limited role in the remainder of this book and the literature provides many very good textbook treatments (see the bibliographic notes at the end of this chapter), we will give this subject relatively short shrift here. The discussion will be more than adequate to acquaint you with some of the basic categories of questions addressed and answers given. But it does not provide the depth of treatment or precision that you will find elsewhere, and students not already acquainted with this subject will do well someday to study it more intensively and extensively.

We divide our treatment of the subject between two sections. In this section we take no derivatives. In the following section, we take derivatives without end.

The consumer's problem

The theory of Marshallian demand concerns a consumer with preferences as described in section 2.1, who seeks to buy a bundle of goods for consumption. We assume that the space of consumption bundles, X, is the positive orthant of R^K for some positive integer K, with the usual interpretation that K is the number of commodities. Stated in words, then, the *consumer's problem* is

Choose the consumption bundle x that is best according to preferences, subject to the constraint that total cost of x is no greater than the consumer's income.

We can express the constraint part of this fairly easily. We use p from R^K to denote the price vector; that is, $p = (p_1, \ldots, p_K)$, where p_j is the price of one unit of good j. We assume that the consumer has a fixed amount of money to spend on his[5] consumption bundle: Let Y denote the consumer's *income*. Then we write the constraint as $\sum_{j=1}^{K} p_j x_j \leq Y$, or, for short, $p \cdot x \leq Y$, where $\cdot$ is the dot or scalar product.[6]

It is implicit in this formulation that the consumer's choice of amounts to consume does not change unit prices of the goods; the consumer faces a *linear price schedule* (no quantity discounts, no increases in price as he tries to buy more, etc.). This is a standard assumption in microeconomics. Quantity discounts are ruled out because, if they were significant, a consumer could buy a lot of the good at a discount and then resell it to other consumers. (But what if resale were impossible? We'll discuss this sort of thing in chapter 9.) And the rationale for the assumption that prices do not increase with increased demand by the single consumer is that the consumer's demand is only a small part of all the demand for the good. This assumption of linear prices is not necessary for a lot of the theory, but it is convenient, and we'll hold to it throughout this chapter.

[5] Generic consumers in this book will change gender occasionally — not so much, I hope, that confusion is created, but often enough so that I am somewhat evenhanded. In particular, later in the book when we have numbered individuals, all odd numbered individuals will be women and all even numbered individuals will be men.

[6] The use of the term "income" is traditional, although "wealth" or "resources" might, for some applications, give more appropriate connotations. The use of an uppercase Y is meant to prevent confusion with consumption bundles; we will sometimes use y as an alternative to x.

That takes care of representing the budget constraint. We can rewrite the consumer's problem as

Choose that consumption bundle x that is best according to preferences, subject to $p \cdot x \leq Y$.

Not bad, but still not great. We still have the objective ("best according to preferences") with which to contend. But reaching back to the previous section, we assume that our consumer has preferences that are asymmetric, negatively transitive, and continuous. This implies that there is a continuous numerical representation U for the consumer's preferences,[7] and we can write the consumer's problem as

$$\textit{Choose } x \textit{ to maximize } U(x) \textit{ subject to } p \cdot x \leq Y \text{ and } x \geq 0. \qquad \text{(CP)}$$

(The $x \geq 0$ part comes from our assumption that the set of feasible consumption bundles, or X, is the positive orthant in R^K.) If we assume that U is well behaved, concave, say, and differentiable, we can even begin creating Lagrangians, differentiating, and so on.

Before making any further assumptions, however, we will see what we can get out of the assumptions that the consumer's preferences are asymmetric, negatively transitive and continuous or, equivalently, that there is a continuous function U that represents preferences.

Right away, we have to wonder whether (CP) has a solution at all.

Proposition 2.11. *If preferences are asymmetric, negatively transitive, and continuous, or equivalently if they are represented by a continuous function U, then:*

(a) The problem (CP) has at least one solution for all strictly positive prices and nonnegative levels of income.

(b) If x is a solution of (CP) for given p and Y, then x is also a solution for $(\lambda p, \lambda Y)$, for any positive scalar λ.

(c) If in addition to our basic three assumptions on preferences, we assume as well that preferences are convex, then the set of solutions to (CP) for given p and Y is a convex set. If preferences are strictly convex, then (CP) has a unique solution.

(d) If in addition (to the original three assumptions) preferences are locally insatiable and if x is a solution to (CP) at (p, Y), then $p \cdot x = Y$.

Let us comment on these four assertions in turn. Before doing so, we give one very useful piece of terminology: For prices p and income Y, the

[7] For those readers who skip the smaller type, take my word for it.

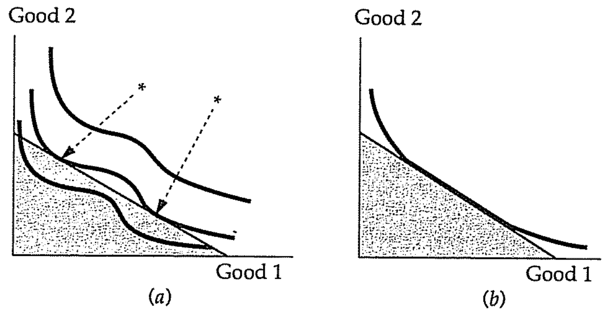

Figure 2.2. Two examples of multiple solutions of (CP).

set of points $\{x \in R^K : p \cdot x \leq Y, x \geq 0\}$ is called the *budget set* for the consumer.

First, the proof of (a) requires a bit of mathematics: You need to know that a continuous function on a compact set achieves a maximum. And then you must show that if prices are strictly positive and income is nonnegative, the budget set is compact. If you know about compact sets, this will be obvious; otherwise, you will have to take (a) on faith.[k]

Part (a) says that (CP) has at least one solution for every p and Y, but there is no reason to suppose that (CP) doesn't have many solutions. In figure 2.2 we show a shaded budget set and preferences by means of the usual indifference curve diagram. In figure 2.2(a) we have nonconvex preferences and two solutions to (CP) marked with stars. In figure 2.2(b) we have convex preferences that are not strictly convex, and we see an entire line segment of solutions.[l]

Part (b) should be immediately obvious because the budget set is unchanged if prices and wealth are scaled up or down proportionately. If (CP) has a unique solution, and we wrote this solution as $x(p, Y)$, then part (b) could be written: $x(p, Y) = x(\lambda p, \lambda Y)$, for $\lambda > 0$. That is, *Marshallian demand* (the function $x(p, Y)$) *is homogeneous of degree zero.*

[k] If you know about compact sets, you may also know that it would be enough for U to be upper semi-continuous. In fact, one can define upper semi-continuous preferences and use this assumption in place of our assumption of continuous preferences in most of the development to come. Also, we will not worry here about what happens when prices of individual goods can be zero, but see the attention we pay this issue in chapter 6.

[l] The preferences represented in 2.2(b) are semi-strictly convex, so that property won't help with multiple solutions. Note also that in figure 2.2 we show only monotone preferences. Draw pictures with preferences that aren't monotone and, especially for part (d) of the proposition, with preferences that aren't locally insatiable.

Part (c) of the proposition is easy to prove: Budget sets are convex: If x and x' are two solutions of (CP) for given p and Y, then $ax + (1-a)x'$ for $a \in [0, 1]$ is in the budget set. And if preferences are convex, $ax + (1-a)x'$ is at least as good as the worser of x and x'. Since these are (presumed to be) both solutions of (CP), they are equally good and as good as anything in the budget set. But then $ax + (1-a)x'$ is a solution of (CP) as well. And if preferences are strictly convex and there are two distinct solutions x and x', $.5x + .5x'$ would be both strictly better than x and x' and would still be in the budget set, a contradiction.

Turning to (d), we begin by noting that if x is any solution to (CP), then $p \cdot x \leq Y$. The content of (d) lies in the assertion that this weak inequality is in fact an equality. The inequality $p \cdot x \leq Y$ is known as *Walras' law*, whereas the equation $p \cdot x = Y$ is called *Walras' law with equality*: Hence we can paraphrase (d) as: *If the consumer is locally insatiable, Walras' law holds with equality.*[8] The proof runs as follows. If some x that solved (CP) for given p and Y satisfied $p \cdot x < Y$, then the consumer can afford any bundle x' that is in some small neighborhood of x, where the size of the neighborhood has to do with how much slack is in the budget constraint and how large the largest price is. But by local insatiability, in every neighborhood of any point x something is strictly preferred to x. Hence x could not solve (CP).[m]

GARP

The set of solutions of (CP) plays, in the specific context of consumer demand, the role that the choice function c plays in abstract choice theory: It is the observable element of choice. Just as we asked before whether a choice function c was consistent with choice according to some underlying preferences, we ask now: *Is observed demand consistent with preference maximization for some set of preferences?* Note two changes from before: The sets A out of which choice is being made are now budget sets and so are infinite. And even if we know all the solutions of (CP) for all prices p and incomes Y, we don't know choice out of all subsets of X; we only know

[8] The term *Walras' law* is sometimes used only for aggregate demand. So you may encounter treatments of the theory of demand where (d) would be paraphased: The consumer's *budget constraint* is satisfied with equality.

[m] Mathematically inclined readers should supply the formal details. Readers sometimes worry about the possibility of getting stuck along an axis in this proof; while the budget constraint may be slack, perhaps it is one of the nonnegativity constraints that binds and keeps the consumer from attaining higher utility. But note carefully that local insatiability is defined relative to the nonnegativity constraints. It says that *within* X and near to every x is something that is strictly better than x. So we can't get stuck along one of the axes.

choice out of budget sets. Since we don't know "c" for all subsets, it may be harder to answer the question posed above.

But, since much of economics is built on the assumption that consumers act as in the problem (CP), it is natural to look at the observed demand and devise and run tests that could lead us to reject this model. That is, it is natural to try to answer the question above. This has been the object of much attention in the theory of the consumer, and two different sorts of answers come out. We give the first here; the second, which is very different in style, will be given at the end of the next section (2.3).

For the balance of this subsection, the assumption is that we observe a finite set of demand data. That is, for a finite collection of prices and income levels $(p^1, Y^1), (p^2, Y^2), \ldots, (p^n, Y^n)$, we observe corresponding demands $x^1, \ldots, x^n$ by our consumer. We do *not* preclude the possibility that at prices p^i and income Y^i there are solutions to (CP) other than x^i. And we set out to answer the question: Given this finite set of data, are the choices we observe consistent with the standard model of some locally insatiable consumer solving (CP)?

The answer is somewhat like the revealed preference axiom we discussed in section 2.1. First, if x is chosen at (p, Y) and $p \cdot x' \leq Y$, then we would have to conclude that $x \succeq x'$ because x is chosen when x' is available. Assuming local insatiability, if the same were true with $p \cdot x' < Y$, then we would know that $x \succ x'$, because otherwise something close to x' would be strictly better than x' and still feasible at prices p with income Y, and x was chosen over this. We can draw pictures of this. In figure 2.3, we show the budget set defined by $p \cdot x \leq Y$ and $x \geq 0$. Suppose the point marked x is chosen. Since everything in the budget set (the shaded

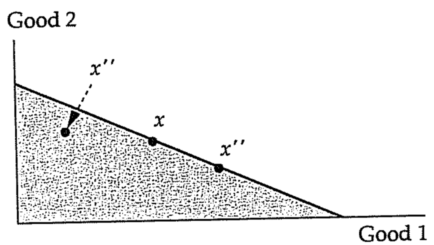

Figure 2.3. Revealed preference from demand data.
If the point x is demanded from the shaded budget set, then x is revealed to be at least as good as points such as x' and, assuming the consumer is locally insatiable, strictly better than points such as x''.

region) was feasible, x must be at least as good as those. In particular, x must be at least as good as x'. And if the consumer is locally insatiable, x must be strictly better than the point marked x''.

Now suppose that this sort of consideration "revealed" that $x^1 \succ x^2 \succeq x^3 \succeq x^4 \succeq x^1$ for some chain of bundles x^1, x^2, x^3, and x^4. We would clearly be unable to rationalize this with preferences that had the properties we've been assuming; a numerical representation would be impossible since we would need to have $U(x^1) > U(x^2) \geq U(x^3) \geq U(x^4) \geq U(x^1)$. This leads to the so-called *Generalized Axiom of Revealed Preference* or GARP, as follows:

Definitions. *Take any finite set of demand data: x^1 chosen at (p^1, Y^1), x^2 chosen at (p^2, Y^2), ..., and x^n chosen at (p^n, Y^n). If $p^j \cdot x^i \leq Y^j$, the data are said to reveal that x^j* ***is weakly preferred to*** *x^i, written $x^j \succeq x^i$. And the data reveal that x^j* ***is strictly preferred to*** *x^i, written $x^j \succ x^i$, if $p^j \cdot x^i < Y^j$. The data are said to satisfy GARP if, for preferences (weak or strong) that the data reveal, you cannot construct a cycle $x^{n_1} \succeq x^{n_2} \succeq \ldots \succeq x^{n_1}$, where one or more of the $\succeq$s is a $\succ$.*

Proposition 2.12. *A finite set of demand data satisfies GARP if and only if these data are consistent with maximization of locally insatiable preferences.*

The argument just sketched shows that satisfaction of GARP is necessary for there to be underlying well-behaved preferences. Sufficiency of GARP is harder to show and is perhaps surprising, but nonetheless, for a finite collection of demand data, if GARP is satisfied, you cannot reject the model of the consumer we have posed.[n]

This is all quite abstract, so let's have an example. Suppose our consumer lives in a three-commodity world.

When prices are $(10, 10, 10)$ and income is 300, the consumer chooses the consumption bundle $(10, 10, 10)$.

When prices are $(10, 1, 2)$ and income is 130, the consumer chooses the consumption bundle $(9, 25, 7.5)$.

When prices are $(1, 1, 10)$ and income is 110, the consumer chooses the consumption bundle $(15, 5, 9)$.

[n] If you are so minded, try a proof. Can you show that these preferences can be assumed to be continuous? Can you show that these preferences can be assumed to be convex? That is, are more testable restrictions generated from the assumptions that preferences are continuous and/or convex? What about strict convexity? If you try this, compare whatever you get with problem 4.

Prices \ Bundles	(10,10,10)	(9,25,7.5)	(15,5,9)
(10,10,10)	300	415	290
(10,1,2)	130	130	173
(1,1,10)	120	109	110

Table 2.1. Cost of three bundles at three sets of prices.

From these data, we calculate the cost of each bundle that is selected at each of the three sets of prices. This is done for you in table 2.1. In each case, the bundle selected exhausts the income of the consumer; this is as it should be, given local insatiability. The important things to note are:

When $(10,10,10)$ was chosen (at prices $(10,10,10)$ and income 300), the bundle $(15,5,9)$ could have been purchased with some money left over. Apparently, this consumer strictly prefers $(10,10,10)$ to $(15,5,9)$.

At the second set of prices $(10,1,2)$, since $(10,10,10)$ and $(9,25,7.5)$ both cost 130 and $(9,25,7.5)$ was selected, the latter must be at least as good as $(10,10,10)$.

And at the third set of prices $(1,1,10)$, the bundle $(9,25,7.5)$ costs 109, while $(15,5,9)$ costs 110. And we are told that with income 110, the consumer chose $(15,5,9)$. Hence $(15,5,9) \succ (9,25,7.5)$.

Oops! The data tell us that $(10,10,10) \succ (15,5,9) \succ (9,25,7.5) \succeq (10,10,10)$. Hence these data are inconsistent with consumer behavior based on the standard preference-maximization model. On the other hand, suppose the third piece of data that we have wasn't as above, but was instead:

At prices $(1,2,10)$ and income 115, the bundle selected is $(15,5,9)$.

Then we would come to no negative conclusions. At the first set of prices and income, the bundles $(10,10,10)$ and $(15,5,9)$ are affordable, and as the first bundle is selected, it is revealed (strictly) preferred to the second. At the second set of prices and income level, $(10,10,10)$ and $(9,25,7.5)$ are affordable and the second is selected, so it is revealed to be (weakly) preferred to the first. This is just as before. But now, at the third set of prices and income level, only $(15,5,9)$ (of the three bundles) is affordable. Knowing that it is selected tells us nothing about how it ranks compared to the other two; it could well come at the bottom of the heap. And we know from the result claimed above, because we observe no violations of GARP in these data, the data can indeed be reconciled with a preference ranking of the usual sort on all bundles.

The indirect utility function

We write $\nu(p,Y)$ for the *value of the problem* (CP). That is,

$$\nu(p,Y) = \max\{U(x) : p \cdot x \leq Y \text{ and } x \geq 0\}.$$

In words, the function ν, which is called the *indirect utility function*, says how much utility the consumer receives at his optimal choice(s) at prices p and income Y. Of course the definition of ν does not depend on (CP) having a unique solution. Note also that the units of ν depend on the specific numerical representation U that is used. If we rescale U, say by replacing U with $V(\cdot) \equiv f(U(\cdot))$, then we transform ν by the same rescaling function f. Put another way, we can never observe ν, since its range is something entirely arbitrary and intangible.

Proposition 2.13. *Assume that U is a continuous function that represents locally insatiable preferences. The indirect utility function ν is*

(a) homogeneous of degree zero in p and Y;

(b) continuous in p and Y (for $p > 0$ and $Y \geq 0$);

(c) strictly increasing in Y and nonincreasing in p; and

(d) quasi-convex in (p,Y).

Parts of the proof. Part (a) should be obvious. The technique of proof used for part (b) reappears frequently in microeconomic theory; so if you know enough of the required mathematics, pay close attention. We take two steps:

Step 1. Suppose that (p^n, Y^n) is a sequence with (bounded) limit (p,Y) such that $p > 0$. Let x^n be a solution of (CP) at (p^n, Y^n) so that $\nu(p^n,Y^n) = U(x^n)$, and let n' be a subsequence along which $\lim_{n'} U(x^{n'}) = \limsup_n \nu(p^n,Y^n)$. Since $p > 0$, one can show that (for large enough n') the union of the budget sets defined by the $(p^{n'}, Y^{n'})$ and (p,Y) is bounded. Thus the sequence $x^{n'}$ lives in a compact space and has some limit point — call it x. Since $p^{n'} \cdot x^{n'} \leq Y^{n'}$, by continuity we know that $p \cdot x \leq Y$. Thus $\nu(p,Y) \geq U(x) = \lim_{n'} U(x^{n'}) = \limsup_n \nu(p^n,Y^n)$.

Step 2. Now let x represent a solution of (CP) for (p,Y), so that $\nu(p,Y) = U(x)$. We know from local insatiability that $p \cdot x = Y$. Let a^n be the scalar

$Y^n/(p^n \cdot x)$. By continuity, $\lim_n a^n = Y/(p \cdot x) = 1$. Thus by continuity of U, $\lim_n U(a^n x) = U(x)$. At the same time, $p^n \cdot a^n x = Y^n$, so that $a^n x$ is feasible for the problem defined by (p^n, Y^n). Hence $\nu(p^n, Y^n) \geq U(a^n x)$, and $\liminf_n \nu(p^n, Y^n) \geq \lim_n U(a^n x) = U(x) = \nu(p, Y)$.

Now combining steps 1 and 2 we see that $\liminf_n \nu(p^n, Y^n) \geq \nu(p, Y) \geq \limsup_n \nu(p^n, Y^n)$, which establishes that $\lim_n \nu(p^n, Y^n)$ exists and is equal to $\nu(p, Y)$.

Part (c) is left for you to do. For the first half of part (c), remember that our consumer is assumed to be locally insatiable. (What would it take to get the conclusion that ν is strictly decreasing in p?)

For part (d), fix some (p^i, Y^i) for $i = 1, 2$ and some $a \in [0, 1]$. Let x be a solution of (CP) at $(ap^1 + (1-a)p^2, aY^1 + (1-a)Y^2)$. We claim that x is feasible for either (p^1, Y^1) or for (p^2, Y^2) (or both); nonnegativity constraints can't be a problem, and if $p^1 \cdot x > Y^1$ and $p^2 \cdot x > Y^2$, then $(ap^1 + (1-a)p^2) \cdot x > aY^1 + (1-a)Y^2$, which contradicts the assumption that x is a solution at $(ap^1 + (1-a)p^2, aY^1 + (1-a)Y^2)$. If x is feasible at (p^1, Y^1), then $\nu(ap^1 + (1-a)p^2, aY^1 + (1-a)Y^2) = U(x) \leq \nu(p^1, Y^1)$, while if x is feasible at (p^2, Y^2), we would conclude that $\nu(ap^1 + (1-a)p^2, aY^1 + (1-a)Y^2) \leq \nu(p^2, Y^2)$. One or the other must be true, so that

$$\nu(ap^1 + (1-a)p^2, aY^1 + (1-a)Y^2) \leq \max\{\nu(p^1, Y^1), \nu(p^2, Y^2)\},$$

which is quasi-convexity.

You may be asking: So what? But we will get something out of the definition of ν and all this work in the next section and in the next chapter.

The dual consumer problem and the expenditure function

To continue preparations for the next section (while continuing to refrain from taking derivatives), we examine the following optimization problem for given utility function $U(\cdot)$, prices p and real number u:

$$\text{minimize } \; p \cdot x \; \text{ subject to } \; U(x) \geq u, x \geq 0. \qquad \text{(DCP)}$$

In words, we seek the minimum amount the consumer must spend at prices p to get for himself utility level u. The appropriate picture is in figure 2.4. The shaded area is the set of bundles x satisfying $U(x) \geq u$; this is the set of points along and above the $U(x) = u$ indifference curve. Iso-expense lines are drawn in; these are lines of the form $p \cdot x =$ constant, for various constants. Note that the closer these lines are to the origin, the

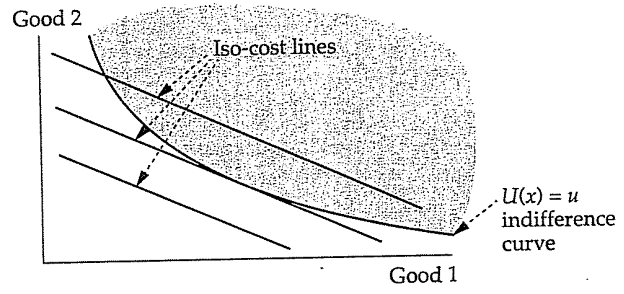

Figure 2.4. The picture for the problem (DCP).

less the constant expense they represent. And the point marked x in the figure is the solution of the problem (DCP).[o]

The problem (DCP) is parametrized by prices p and the utility level u that is to be achieved. (It is also parametrized by the utility function $U(\cdot)$, but we will hold that fixed throughout all our discussions.) As we vary p and u, we get other solutions. We will write $e(p, u)$ for the value of the problem as a function of (p, u); that is,

$$e(p, u) = \min\{p \cdot x : U(x) \geq u, x \geq 0\},$$

and we will call the function e the *expenditure function*.

In parallel with propositions 2.11 and 2.13, we have

Proposition 2.14. *Suppose that p is strictly positive and u is a level of utility that is achievable and greater or equal the utility of consuming the zero vector; that is, $u = U(x)$ for some $x \geq 0$, and $u \geq U(0)$. Then,*

(a) The problem (DCP) has a solution at (p, u). If x solves (DCP) for (p, u), then $U(x) = u$. If U represents convex preferences, the set of solutions to (DCP) is convex. If U represents strictly convex preferences, (DCP) has a single solution for each (p, u). If x solves (DCP) for (p, u), then x is a solution of (DCP) for $(\lambda p, u)$, where λ is any strictly positive scalar.

(b) The expenditure function $e(p, u)$ is homogeneous of degree one in p, or

[o] For reasons we will not explain in this book, this problem is called the *dual consumer problem* (giving the mnemonic [DCP]), because it is dual to the problem (CP) in a precise mathematical sense.

$e(\lambda p, u) = \lambda e(p,u)$ for $\lambda > 0$. The expenditure function is strictly increasing in u and nonincreasing in p.

(c) The expenditure function is concave in p.

Proofs of parts (a) and (b) are left to you with a few remarks and hints. Recall that we needed local insatiability to prove that Walras' law held with equality at any solution of (CP). But here we don't need local insatiability to show that $U(x) = u$ at a solution x. (In fact, we don't need to assume local insatiability in this proposition at all! But we do assume that U is a continuous function.) So how does the argument go? Suppose that x solves (DCP) for (p,u) with $U(x) > u$. If $u > U(0)$, our consumer is spending something at x to achieve u. But then consider bundles of the form αx for α a bit less than one. These bundles cost proportionately less than x and, by continuity of U, for α close enough to one they give more utility than u (since $U(x) > u$). This would contradict the assumption that x solves (DCP). And if $u = U(0)$, the solution to (DCP) at (p,u) is clearly $x = 0$.[p]

Part (c) of the proposition is crucial to later events, so we give the proof here. Let x solve (DCP) for $(ap + (1-a)p', u)$, so that $e(ap + (1-a)p', u) = (ap + (1-a)p') \cdot x$. Since $U(x) = u$, the bundle x is always a feasible way to achieve utility level u, although it may not be the cheapest way at prices other than $ap + (1-a)p'$. Accordingly,

$$e(p,u) \le p \cdot x \text{ and } e(p',u) \le p' \cdot x.$$

Combining these two inequalities gives

$$\begin{aligned} ae(p,u) + (1-a)e(p',u) &\le ap \cdot x + (1-a)p' \cdot x \\ &= (ap + (1-a)p') \cdot x = e(ap + (1-a)p', u). \end{aligned}$$

That does it.

Comparative statics

We are almost ready to start taking derivatives, but before doing so, we take one further excursion into definitions and basic concepts.

A standard exercise in the theory of consumer demand is to ask how demand responds to changes in various parameters such as prices and income. When (CP) has more than one solution this exercise is a bit hard to interpret, so we assume for the balance of this subsection that (CP) has a unique solution.[9] We write $x(p,Y) = (x_1(p,Y), \ldots, x_K(p,Y))$ for the solution; that is, $x_j(p,Y)$ is the amount of good j purchased by our consumer when he faces prices p and has income Y to spend.

[p] What would happen if $u < U(0)$?

We are interested in how $x(p,Y)$ (and its constituent components) change with changes in the various p_i and in Y.

(a) For example, how does $x_j(p,Y)$ change with changes in p_j? If good j becomes more expensive, we "expect" that less j will be demanded; in fact, we expect this so much that a good for which this is true is called a *normal* good. When the demand for good j rises with increases in the price of good j, we say that good j is a *Giffen good.* We will return briefly to normal and Giffen goods and this general subject of *own price effects* in the next section (2.3).

(b) How does $x_j(p,Y)$ change with changes in p_i for $i \neq j$, holding fixed the prices of other goods and income? This depends in the first place on the relationship between goods i and j. If good i, say, is popcorn and j is popcorn poppers, a rise in the price of i will probably cause the demand for good j to fall. If good i is popcorn and j is peanuts, demand for j will probably rise with a rise in the price of i. We will have a little to say about this subject of *cross-price effects* in section 2.3, as well.

(c) How does $x_j(p,Y)$ change with changes in income Y? If we think of two goods, 1 and 2, we get pictures like those in figure 2.5; the heavy curve in each case shows how demand shifts with shifts in Y. These curves are called *income expansion paths* or *Engel curves.* In most cases, we expect that x_j will rise with increasing income, but it is possible that demand for j falls with increases in Y; goods j such that x_j falls as Y increases are called *inferior* goods. An example might be potatoes, insofar as an increase in income means the consumer can afford to substitute more expensive, higher "quality" food such as meat for potatoes. Good 1 in figure 2.5(a) is an example of an inferior good, at least for a range of Y. In case x_j increases with increasing Y, we become interested in whether x_j increases

[9] A sufficient condition for this is that preferences are strictly convex; see proposition 2.11. But strict convexity is unpalatable in some cases. For example, I have no use whatsoever for hydrochloric acid. My preferences are completely unaffected by the amount of hydrochloric acid in my consumption bundle. Hence my preferences are not strictly convex along the dimension of hydrochloric acid. On the other hand, this doesn't affect the uniqueness of a solution to (CP) for me, since at positive prices for hydrochloric acid I will buy no hydrochloric acid at all. On this point, see problem 10.

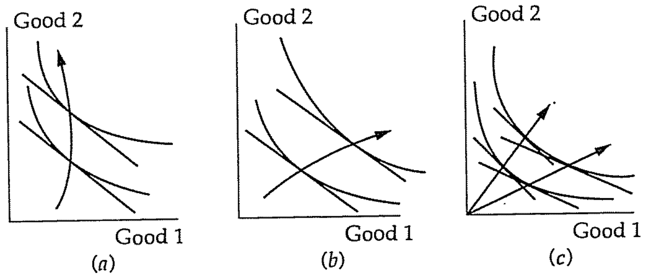

Figure 2.5. Various income expansion paths or Engel curves.
In (a), good 1 is an inferior good over a range of Y. In (b), good 1 is a luxury good and good 2 is a necessary good. In (c), demand is homothetic.

more or less proportionately with Y. When $x_j(p,Y)/Y$ increases (a higher fraction of income is spent on j with increases in Y), j is called a *luxury* good; when $x_j(p,Y)/Y$ decreases with increases in Y, j is called a *necessary* good. In figure 2.5(b), good 1 is a luxury good and good 2 is a necessary good. When $x_j(p,Y)/Y$ doesn't change with changes in Y for all j, the consumer's preferences are said to be *homothetic*, as in figure 2.5(c).

These are examples of *comparative statics* exercises of the simplest kind — asking how consumer behavior changes with changes in underlying parameters. The term "statics" is used because it is assumed that the consumer fully adjusts to the changes in the parameters. Indeed, we might better phrase the question: What differences would be observed if we observed two different but identical consumers placed in two different circumstances? More difficult comparative statics exercises, built from these simple exercises, ask how consumer behavior shifts with shifts in other "parameters" of the model. How does demand for cigarettes change if we impose an excise tax on cigarettes? How does demand for cigarettes change if we impose an income tax on consumers?

We should stress that each of the exercises involves shifts in demand as one parameter changes, the others being held fixed. If we assumed that the demand function $x(p,Y)$ were differentiable, we would be asking questions about the partial derivatives of x. In a moment we will assume that $x(p,Y)$ is differentiable, and so when we return to these questions, they will be phrased in terms of partial derivatives. You may find it helpful to recast the terms just given in the language of partial derivatives before reading ahead.

2.3. Marshallian demand with derivatives

The first-order conditions for (CP)

How do we solve the problem (CP)? The standard approach is to assume that the utility function U is differentiable; form a Lagrangian; and look at the combined first-order conditions and complementary slackness conditions. (Consult appendix 1 immediately if you have no idea what this means, or if you need a quick review.) Letting λ be the multiplier on the budget constraint $p \cdot x \leq Y$ and μ_j (for $j = 1, \ldots, K$) the multiplier on the constraint $x_j \geq 0$, the Lagrangian is

$$U(x) + \lambda\left(Y - \sum_{j=1}^{K} p_j x_j\right) + \sum_{j=1}^{K} \mu_j x_j,$$

and the first-order conditions read

$$\frac{\partial U}{\partial x_j} = \lambda p_j - \mu_j.$$

The multipliers are all constrained to be nonnegative and, of course, the solution must obey the original constraints $p \cdot x \leq Y$ and $x \geq 0$. The complementary slackness conditions must also hold:

$$\lambda(Y - p \cdot x) = 0 \quad \text{and} \quad \mu_j x_j = 0 \text{ for } j = 1, \ldots, K.$$

We can rewrite this as follows: Since the μ_j are nonnegative, these multipliers can be eliminated and the first-order condition for x_j and the complementary slackness condition for μ_j can be combined succinctly as

$$\frac{\partial U}{\partial x_j} \leq \lambda p_j, \text{ with equality if } x_j > 0.$$

(Be absolutely certain that you see why this is equivalent to the combined first-order condition with the μ_j included and the complementary slackness condition on μ_j.) Assuming prices are all strictly positive (which we do throughout this chapter), we can rewrite again as

$$\frac{1}{p_j}\frac{\partial U}{\partial x_j} \leq \lambda, \text{ with equality if } x_j > 0.$$

Or in words, for goods that are consumed in positive levels, the ratios of the marginal utility of the goods to their respective prices must be equal, and these ratios are greater than the corresponding ratios for goods that are not consumed.

If we know that $\lambda > 0$, we can rewrite in yet another form that you may recall from intermediate micro or even from principles courses. For two goods i and j that are consumed in positive amounts at the optimum,

$$\frac{\partial U}{\partial x_i} \Big/ \frac{\partial U}{\partial x_j} = p_i/p_j.$$

Or, in words, the ratio of marginal utilities equals the ratio of prices. You may recall this stated as *the marginal rate of substitution of good i for good j (along an indifference curve) equals the ratio of their prices.* In fact, even if $\lambda = 0$ we have this relationship, as long as we recognize that when $\lambda = 0$, the first-order conditions read that $\partial U/\partial x_i = 0$ for goods that are consumed in positive amounts, and we interpret $0/0$ as being any number we wish.

But is $\lambda > 0$? Because we will assume that it is throughout the remainder of this section, some justification should be given. Recall that the indirect utility function $\nu(p, Y)$ gives the value of the problem (CP) as a function of the parameters p and Y. If you have a good background in constrained optimization, you will note immediately that λ is $\partial \nu/\partial Y$. (Assume that ν is differentiable for the time being; we'll discuss this later.) Proposition 2.13 shows that ν is strictly increasing in Y, assuming local insatiability, so things look good. But there are strictly increasing, differentiable functions whose derivatives are zero at isolated points. And so it is with $\partial \nu/\partial Y$ and λ; depending on the representation U of preferences that we use, we can produce examples where the multiplier is sometimes zero. This is true even if U represents convex preferences, as long as we only insist on U being quasi-concave; there are strictly increasing and quasi-concave functions whose derivatives go to zero at points. Now if U is a concave function, we are certainly in business. Then we can show that $\nu(p, Y)$ is concave in Y, and a strictly increasing, concave function can never have zero derivative. But assuming concavity of U is hard to do on first principles (until next chapter). Conclusions: In any application where you specify a concave function U, you can freely assume and will always find that the multiplier λ is strictly positive. But if you assume that U is quasi-concave based on an assumption of even strictly increasing and strictly convex preferences, it takes a little more to be sure that the multiplier is strictly positive.

We have all this from writing down the first-order and complementary slackness conditions. But what is their standing in relation to the problem (CP)? If you find a solution to (CP), must these conditions hold (for some multipliers)? If you find a solution to the first-order conditions and the complementary slackness conditions (that satisfies the sign constraints on the multipliers and the various feasibility constraints on the variables x_i), will it be a solution to (CP)? Those readers who have studied constrained optimization will know that for this problem, *if U is concave, these conditions are necessary and sufficient for a solution.* Those readers who have not studied constrained optimization now have a bit more incentive to do so.

But what is the status of an assumption that U is concave? And all this is predicated on the assumption that U is differentiable; can that assumption be grounded in some assumption on preferences $\succ$? I do not know of any natural conditions on $\succ$ that would guarantee that $\succ$ admits a differentiable representation. Note well the way this was put. We have no hope whatsoever that every representation of a given $\succ$ will be differentiable, since we already saw that some representations of $\succ$ will not be continuous. It is the same, as we've noted, for concavity. Concavity of U will not necessarily be preserved by monotonic transformations, so we can't say anything that will make every representation of $\succ$ concave. As we saw at the end of section 2.1, the "natural" notion of convexity of $\succ$ will only guarantee that $\succ$ has a quasi-concave representation. And if you are able to do problem 7, you will discover that otherwise well-behaved convex preferences may admit no concave representations at all. So, granting differentiability, we can still ask for the status of the first-order and complementary slackness conditions for a quasi-concave U. These are necessary, but not quite sufficient for optimality; those of you who have studied constrained optimization now have an incentive to go back and review those chapters at the end of your old textbook about quasi-concavity (or -convexity).

Assumptions galore

Now we make some serious assumptions. We henceforth assume that both (CP) and (DCP) have unique solutions for every (p, Y) and (p, u). As noted, strict convexity of preferences suffices for this but is not an innocuous assumption. We write $x(p, Y)$ for the solution of (CP) as a function of (p, Y) and $h(p, u)$ for the solution of (DCP) as a function of (p, u). The function $x(p, Y)$ is called the *Marshallian* demand function, and $h(p, u)$ is called the *Hicksian* demand function.

We further assume that Marshallian demand, Hicksian demand, the indirect utility function and the expenditure function are all continuously differentiable functions of all their arguments.

What justification can there be for these assumptions? If you followed the proof of proposition 2.13(b) (continuity of the indirect utility function), you should be able to use that technique with the assumptions that (CP) and (DCP) have unique solutions to show that the four functions are all continuous. But differentiability is quite a bit more. We do not provide justification here. In more advanced treatments of the subject, you will find that, in essence, one requires that U is twice-continuously differentiable (and well behaved along axes.) And as we had no reason to suppose from first principles concerning $\succ$ that U is differentiable, we have even less reason to assume that it is twice differentiable. You can also ask how much of what follows can be extended to cases where our assumptions of differentiability are not met. All these matters we simply sweep under the rug, and they constitute one of the principal reasons that the treatnent of demand theory here is just a sketch.

Hicksian demand and the expenditure function

Proposition 2.15. *The expenditure function and the Hicksian demand function are related as follows:*

$$h_i(p,u) = \frac{\partial e(p,u)}{\partial p_i}.$$

Proof. We give a very slick graphical proof of this result first. Fix the utility argument u^* and all the prices p_j^* except for p_i, and graph the function

$$p_i \to e((p_1^*,\ldots,p_{i-1}^*,p_i,p_{i+1}^*,\ldots,p_K^*),u^*).$$

We are assuming that this function is differentiable.[q] Now at the value $p_i = p_i^*$, what is its derivative? Since we know that we can get utility u^* from the bundle $h(p^*,u^*)$, we know that

$$e((p_1^*,\ldots,p_{i-1}^*,p_i,p_{i+1}^*,\ldots,p_K^*),u^*) \le p_i h_i(p^*,u^*) + \sum_{j\neq i} p_j^* h_j(p^*,u^*).$$

The function on the right-hand side, therefore, is everywhere above the function $e(p,u^*)$, and it touches at p^*, so it must be tangent as shown in figure 2.6. But it is a linear function of p_i, and its slope (in the p_i direction) is $h_i(p^*,u^*)$. Done.[r]

[q] In fact, we know from proposition 2.14 that this is a concave function, which almost, but not quite, gives differentiability; see the fn. following.

[r] (1) What if we didn't assume that $e(p,u)$ is differentiable? It is still concave in p_i, so we would get a similar picture, except that we might find that e has a kink at p_i^*. We would then get a statement about how $h_i(p^*,u^*)$ lies between the right- and left-hand partial derivatives of e, where their existence is guaranteed by concavity. (2) Suppose that we assume that e is differentiable, but we don't assume directly that (DCP) has a unique solution. What does the proposition tell us?

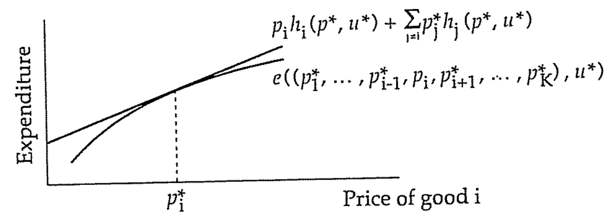

Figure 2.6. Connecting Hicksian demand and the expenditure function.

This proof is so slick that you may be unable to replicate it in other situations. So we will give a more dreary algebraic proof, which illustrates a technique that is much used in this branch of microeconomics. As a first step, differentiate both sides of the accounting identity $e(p,u) = p \cdot h(p,u)$ with respect to p_i. This gives

$$\frac{\partial e}{\partial p_i} = h_i(p,u) + \sum_{j=1}^{k} p_j \frac{\partial h_j}{\partial p_i}.$$

In words, if we raise the price of good i, then the resulting change in expenditure needed to reach utility level u comes from two terms. First, the amount h_i of good i that we were buying before is more expensive; expenditure rises at precisely the rate $h_i(p,u)$. And, second, we'll probably buy less i and more or less of other things; the sum term above gives the "cost" of changes in the optimal bundle to buy. The result we are supposed to be heading for says that the summation term is identically zero.

To see why this is, recall that $h(p,u)$ is the solution of (DCP), or

$$\min_x p \cdot x \quad \text{subject to} \quad U(x) \ge u.$$

If we make η the Lagrange multiplier on the constraint, the first-order condition (for x_i) reads

$$p_i = \eta \frac{\partial U}{\partial x_i},$$

evaluated, of course, at the optimum, which is $h(p,u)$. (We ignore nonnegativity constraints on x here; the diligent reader may wish to replicate our argument with that additional complication.) This has a partial of U with respect to x_i, whereas we are interested in getting rid of a sum involving partials of h with respect to p_i; so we aren't quite there yet. But now a little sleight of hand comes into play. First write the accounting identity

$$U(h(p,u)) = u.$$

Implicitly partially differentiate both sides with respect to p_i, and you get

$$\sum_{j=1}^{K} \frac{\partial U}{\partial x_j}\frac{\partial h_j}{\partial p_i} = 0.$$

We can push our first-order conditions into this, and get

$$(1/\eta)\sum_{j=1}^{K} p_j \frac{\partial h_j}{\partial p_i} = 0.$$

This is just what we want, as long as the Lagrange multiplier η isn't zero or infinite. There are ways to ensure this. See the discussion above concerning the multiplier on the budget constraint in (CP).

Roughly put, our technique here is to substitute into one equation a first-order condition for something that is optimal in a constrained optimization problem. This technique is closely connected to the picture we drew: In essence, we are trading on the fact that, at the optimum, when the price of p_i changes, we have the same marginal impact on expenditure by "absorbing" all the change into good i as we would by fully reoptimizing. If we fully reoptimized our consumption bundle, we would do better than by absorbing the change into good i alone; this is why the expenditure function lies everywhere below the line in figure 2.6. But because, at a solution of (DCP), the ratios of marginal utilities to prices of the goods are equal, the difference on the margin between fully reoptimizing and using good i alone is of second order (assuming differentiability of the expenditure function and ignoring nonnegativity constraints, which are left for you to handle). This technique is formalized in general in *the envelope theorem*, so-called because the expenditure function is the lower envelope of linear functions like the one we have drawn. If you have not already studied the envelope theorem in general, you may wish to do so (elsewhere).

Roy's identity

Taking the partial derivative of the expenditure function with respect to the ith price gave us Hicksian demand for the ith good. Can we get anything nice by taking partial derivatives of the indirect utility function? An affirmative answer is given by *Roy's identity*.

Proposition 2.16. *Marshallian demand and the indirect utility function are related as follows:*

$$x_i(p,Y) = -\frac{\partial \nu}{\partial p_i}\Big/\frac{\partial \nu}{\partial Y}.$$

Proof. Suppose that $x^* = x(p,Y)$. Let $u^* = U(x^*)$. Then we claim that $x^* = h(p,u^*)$, and $Y = e(p,u^*)$. (This step is left to you. Remember that we are assuming that [CP] and [DCP] have unique solutions.) Thus $u^* \equiv \nu(p, e(p,u^*))$ for fixed u^* and all p. Differentiate this with respect to p_i and you get

$$0 = \frac{\partial \nu}{\partial p_i} + \frac{\partial \nu}{\partial Y}\frac{\partial e}{\partial p_i}.$$

Use proposition 2.15 to replace $\partial e/\partial p_i$ with $h_i(p,u^*) = x_i^* = x_i(p,Y)$ and rearrange terms.

Roy's identity may seem a bit mysterious, but it has a fairly intuitive explanation which requires that you know that $\partial \nu/\partial Y$ is the multiplier λ in (CP). Assuming $x_i(p,Y) > 0$, the first-order condition for x_i in (CP) can be written

$$\frac{1}{p_i}\frac{\partial U}{\partial x_i} = \lambda.$$

Roy's identity can be written

$$-\frac{\partial \nu}{\partial p_i} = \frac{\partial \nu}{\partial Y}x_i,$$

so substituting λ for $\partial \nu/\partial Y$ and combining, we get

$$-\frac{\partial \nu}{\partial p_i} = \lambda x_i = \frac{x_i}{p_i}\frac{\partial U}{\partial x_i}. \qquad (♠)$$

(What happens if $x_i = 0$?) What does this mean? Imagine that the price of good i decreases by a penny. How much extra utility can our consumer obtain? The consumer with income Y to spend, if he buys the bundle he was buying before, has x_i pennies left over because good i is cheaper. He can use this "extra" income naively, spending all of it on good i. If he does this, he can buy (approximately) one penny times x_i/p_i more of good i, which raises his utility by one penny times $(x_i/p_i)(\partial U/\partial x_i)$. Now this is a naive response to a decrease in the price of good i; our consumer will probably do some further substituting among the various goods. But, what

Roy's identity shows, these substitutions will *not* have a first-order effect on the consumer's utility; the main effect is captured by naively spending his "windfall" entirely on good i. (You should complement this discussion by obtaining (♠) in two related ways: drawing a picture like figure 2.6 and by constructing an argument where some first-order condition is used as in the previous subsection.)

The Slutsky equation: Connecting Marshallian and Hicksian demand

What happens to Marshallian demand for commodity j starting at prices p and income Y if the price of commodity i rises? Two things: First, in real terms the consumer is a bit poorer; the general "price index" has risen. Roughly, since the consumer was buying $x_i(p,Y)$ of good i, his "real income" falls at approximately the rate $x_i(p,Y)$ (a penny rise in the price of i means $x_i(p,Y)$ pennies less to spend). This will cause our consumer to change his demand for j, at a rate (roughly) $-(\partial x_j/\partial Y)x_i(p,Y)$. That is, we multiply the rate of change in consumption of j per unit change in income times the rate of change in real income that we just computed.

Second, good i no longer looks so attractive — its relative price has risen. So the consumer will (probably) consume less of i. And, depending on the relationship between i and j, this may cause the consumer to take more of j or less. In any event, there is a "cross-substitution" effect on the consumption of j. We want to isolate this substitution effect from the change in consumption of j due to the change in real income, so we look for some sort of compensated demand function — change the price of good i, compensate the consumer so that his real income stays the same, and see how the change in p_i affects the compensated demand for j.

There are two obvious ways to compensate our consumer. We could raise Y enough so that the consumer could afford the bundle that he was previously buying — Slutsky compensation. Or we could give the consumer sufficient income so that, after he optimizes, he is as well off in terms of his preferences as before — Hicks compensation. If you took intermediate micro, or even a good principles course, you probably will remember something like this.

Our friend the Hicksian demand function is based on Hicksian compensation. Actually making Hicksian compensation is not going to be possible because it depends on the unobservable utility function; it is meant solely as a theoretical construct. Despite this, since we have this demand function defined, let's adopt Hicksian compensation. The Hicks compensated substitution term is simply $\partial h_j/\partial p_i$. (Slutsky compensation makes a nice homework problem.)

So we expect that the change in Marshallian demand of j as p_i changes is the sum of the income effect and the compensated substitution effect. Let us record our expectations in the form of a proposition.

Proposition 2.17. *Marshallian demand and Hicksian demand are related as follows:*

$$\frac{\partial x_j}{\partial p_i} = \frac{\partial h_j}{\partial p_i} - \frac{\partial x_j}{\partial Y}x_i, \qquad (\star)$$

where we evaluate, for given prices p and income level Y, the Marshallian demand at those prices and that income, and, for the Hicksian demand function, the utility level achieved at that Marshallian demand point.

Expecting that this equation holds and proving it are two different things. Our verbal argument above is loose in two places: We don't know that Hicksian compensation is correct, and our income adjustment x_i is not quite correct because of the substitution out of x_i. But a formal proof is relatively painless. Write the identity $x_j(p, e(p,u)) \equiv h_j(p,u)$ and differentiate both sides with respect to p_i. We get

$$\frac{\partial x_j}{\partial p_i} + \frac{\partial x_j}{\partial Y}\frac{\partial e}{\partial p_i} = \frac{\partial h_j}{\partial p_i}.$$

Since $\partial e/\partial p_i$ is $h_i(p,u) = x_i(p, e(p,u))$, this reads

$$\frac{\partial x_j}{\partial p_i} + \frac{\partial x_j}{\partial Y}x_i(p, e(p,u)) = \frac{\partial h_j}{\partial p_i},$$

which, once we make right substitutions for dummy variables, is equation $(\star)$. Equation $(\star)$ is known as the *Slutsky equation.*

About differentiable concave functions

(You may be wondering where all this is headed. We are certainly taking lots of derivatives, and it isn't at all clear to what end we are doing so. Some results are coming, but we need a bit more setting-up to get them. Please be patient.)

To motivate what comes next, think of a twice-continuously differentiable concave function of one variable. A concave function (of one variable) is a function whose derivative is nonincreasing; hence it is a function whose second derivative is nonpositive.

For concave functions of several variables this generalizes as follows: For a given twice-continuously differentiable function $f : R^K \to R$ and any point $z \in R^K$, let $H(z)$ be $K \times K$ matrix whose (i, j)th element is the term

$$\left.\frac{\partial^2 f}{\partial z_i \partial z_j}\right|_z .$$

That is, $H(z)$ is the matrix of mixed second partials of f, evaluated at z. This matrix is called the *Hessian matrix* of f. Note that a Hessian matrix is automatically symmetric; the mixed second partial of f with respect to z_i and z_j is the same as the second partial of f with respect to z_j and z_i.

Definition. *A $k \times k$ matrix H is **negative semi-definite** if for all $\zeta \in R^K$, $\zeta H \zeta \le 0$.*

(By $\zeta H \zeta$, we mean $\zeta^T H \zeta$, where the superscript T denotes transpose, and we think of ζ as a $K \times 1$ column vector.)

Corollary of the definition. *If H is a negative semi-definite $K \times K$ matrix, then $H_{ii} \le 0$ for $i = 1, \ldots, K$.*

The corollary follows immediately from the definition, if we take for ζ a vector with 1 in the ith position and 0 in all others.

Mathematical fact. *A twice-continuously differentiable function $f : R^K \to R$ is concave if and only if its Hessian matrix (evaluated at each point in the domain of f) is negative semi-definite.*[8]

The main result, with applications to comparative statics

We can now connect all the pieces and obtain some results. We state the conclusion first, and then give the chain of logic that yields it.

Proposition 2.18. *If $x(p, Y)$ is a Marshallian demand function, then the $K \times K$ matrix whose ijth term is*

$$\frac{\partial x_i(p, Y)}{\partial p_j} + \frac{\partial x_i(p, Y)}{\partial Y} x_j(p, Y),$$

*which is called the matrix of **substitution terms**, is symmetric and negative semi-definite.*

[8] This is true as well if f is defined on an open domain within R^K, which in fact is how we will use it.

This conclusion is reached as follows. By the Slutsky equation, the i, jth term of our matrix is $\partial h_i(p, u)/\partial p_j$, evaluated at $(p, U(x(p, Y)))$. This, according to proposition 2.15, is $\partial^2 e/(\partial p_i \partial p_j)$. So the matrix of substitution terms is the Hessian matrix of the expenditure function. The expenditure function is concave in p by proposition 2.14(c). So the result follows from properties of Hessian matrices of concave functions.

Of what use is this? We return to the comparative statics exercises sketched at the end of the last section. Good j is inferior if $\partial x_j/\partial Y < 0$; it is a necessary good if $0 \le \partial x_j/\partial Y < x_j/Y$; and it is a luxury good if $x_j/Y \le \partial x_j/\partial Y$. It is a normal good if $\partial x_j/\partial p_j < 0$, and it is a Giffen good if $\partial x_j/\partial p_j > 0$.

Consider the possibility of a Giffen good. By the Slutsky equation,

$$\frac{\partial x_j}{\partial p_j} = \frac{\partial h_j}{\partial p_j} - \frac{\partial x_j}{\partial Y} x_j .$$

We know by the corollary from the previous subsection and proposition 2.18 that $\partial h_j/\partial p_j < 0$; that is, the own-price substitution term is always nonpositive. So the only way that a good could be a Giffen good is if it is an inferior good. It has to be sufficiently inferior so that the income effect term $-(\partial x_j/\partial Y)x_j$ overcomes the substitution term $\partial h_j/\partial p_j$. And, roughly speaking, an inferior good stands a better chance of being a Giffen good if it looms larger in the consumer's consumption bundle; note that $\partial x_j/\partial Y$ is multiplied by x_j. So if one is ever to find a Giffen good (and it isn't clear that one has ever been identified) the place to look is at inferior goods that make up a large part of the consumer's consumption bundle.

Or consider cross-price effects, that is, things like $\partial x_j/\partial p_i$. In informal economics, the term *substitutes* is used to describe pairs of goods where an increase in the price of one good causes an increase in the demand of the second, and *complements* is used when an increase in the price of one good causes a decrease in the demand of the other. If we attempt to formalize this in terms of the signs of $\partial x_j/\partial p_i$, we run into the unhappy possibility that $\partial x_j/\partial p_i < 0$ and, at the same time, $\partial x_i/\partial p_j > 0$. The symmetry of the substitution terms precludes this unhappy possibility when, instead, we define substitute and complement pairs of goods in terms of the sign of $\partial h_j/\partial p_i$. And so it has become common to use this definition, thereby admitting the possibility that, say, $\partial x_i/\partial p_j < 0$ even when i and j are formally substitutes, because the income effect term might outweigh the substitution term.

In general, the Slutsky equation allows us to view comparative statics in prices as the sum of an income effect and a substitution effect, where the latter is given by the partial derivatives (in prices) of Hicksian demand. And our proposition gives us some information about the nature of these substitution effects.

Integrability

The big payoff to all our labors comes when we ask the question: Given a function $x(p, Y)$ that is claimed to be a Marshallian demand function, is some locally insatiable, preference-maximizing consumer behind it?

Why would anyone ever ask such a question? In applied (econometric) work, it is sometimes useful to have parametric functional specifications for a consumer's demand function. Now we could write down nice closed form parametrizations for the utility function U, and then solve (CP) to derive Marshallian demand. But for most analytically tractable specifications of U, the Marshallian demand function that results is a mess. It is often more convenient to specify a parametric form for Marshallian demand directly, being sure that this parametrization is analytically tractable. But then, to be sure that one's parametrization plays by the rules of microeconomic theory, one would want to know that the question asked in the previous paragraph is answered affirmatively.

We assume that we have a functional specification of an entire Marshallian demand function which is differentiable. From easy first principles, we know that Marshallian demand should be homogeneous of degree zero and should obey Walras' law with equality. And we now also know that the matrix of substitution terms (which are functions of Marshallian demand) must be symmetric and negative semi-definite. Subject to some technical caveats, these things are *sufficient* as well: Assuming that Marshallian demand obeys Walras' law with equality and that it is homogeneous of degree zero, if the matrix of substitution terms is symmetric and negative semi-definite, then it can be "integrated up" to get the indirect utility function, from which a representative of the direct utility function can be constructed. We won't even attempt to sketch the proof here; serious students will wish to consult a standard text on demand theory to see how it's done.

2.4. *Aggregate demand*

One further subject will come into play in later developments in this book and should be mentioned — *aggregate consumer demand.*

Until now, we have been considering demand by a single consumer. But in much of this book, we will be interested instead in demand by a host of consumers, which, at prices p, will be the sum of their individual

demands given their individual incomes. Our interest will be strong especially because it is hard (not impossible) to obtain data on the demand by a single consumer.

Of course, total demand will shift about as a function of how individual incomes are distributed, even holding total (societal) income fixed. So it makes no sense to speak of aggregate demand as a function of prices and social income.

Even with this caveat, results analogous to the Slutsky restrictions (proposition 2.18) or GARP do not necessarily hold for aggregate demand. Think of GARP for a moment. Imagine a two-person, two-commodity economy in which we fix each person's income at 1,000. At prices $(10, 10)$, we might have one person choose (25,75) and the second choose (75,25). Hence aggregate demand is $(100, 100)$. Suppose that at prices $(15, 5)$, the first chooses $(40, 80)$ and the second $(64, 8)$. Neither individual violates GARP; the first can't afford $(40, 80)$ at prices $(10, 10)$, so we imagine that he has to settle for (25,75) at those prices. And the second can't purchase (75,25) at the prices $(15, 5)$, so we imagine that she must settle for the worse bundle $(64, 8)$ at the second set of prices.

But aggregate demand is (100, 100) in the first case, which is affordable at both sets of prices, and (104, 88) in the second case, which is both affordable at both sets of prices and inside the "social budget set" at the prices where it isn't chosen. Society doesn't obey GARP, even though both individuals do.

So what can we say about aggregate demand, based on the hypothesis that individuals are preference/utility maximizers? Unless we are willing to make strong assumptions about the distribution of preferences or income throughout the economy (everyone has the same homothetic preferences, for example), there is little we can say. Aggregate demand will be homogeneous of degree zero in prices and (everyone's) income. And if all consumers are locally insatiable, Walras' law will hold with equality for the entire economy. But beyond that, almost anything is possible for aggregate demand. In the advanced literature of demand theory, this result is stated and proved formally.

2.5. *Bibliographic notes*

The material in section 2.1, on preferences, choice, and numerical representations, belongs to the subject of abstract choice theory. For a fairly complete treatment of this subject, see Fishburn (1970). Less complete (and, in particular, lacking the proof that a continuous preference relation admits a continuous numerical representation) but perhaps a bit more readable is Kreps (1988).

We mentioned in section 2.1 problems with the standard models of choice concerning framing. If you are interested, a good place to begin is the brief introductory article by Kahneman and Tversky (1984). For a broader treatment and some applications, see the *Journal of Business* symposium issue of 1987. We also mentioned "time of day" problems; this will be further discussed in chapter 4, and references will be given there.

As for standard references in demand theory, Deaton and Muellbauer (1980) and Varian (1984) are good sources for many of the details and topics omitted here. In particular, both cover functional separability of preferences, which is very important to econometric applications. Katzner (1970) is a good source for seeing how to think about differentiability and other mathematical niceties.

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2.6. Problems

■ 1. Let $\succ$ be a binary relation on a set X, and define $\succeq$ as in section 2.1; $x \succeq y$ if it is not the case that $y \succ x$. Show that $\succ$ is asymmetric if and only if $\succeq$ is complete. Show that $\succ$ is negatively transitive if and only if $\succeq$ is transitive. (Hint: For the second part, it may help to know that the statements "A implies B" and "not B implies not A" are logical equivalents. The latter is called the *contrapositive* of the former. We gave negative transitivity in the form *if* $x \succ y$, *then* for any z, either $x \succ z$ or $z \succ y$. What is the contrapositive of this if–then? Now do you see why it is called negative transitivity?)

■ 2. Show that if $\succ$ is asymmetric and negatively transitive, then $\sim$ defined as in the text is reflexive, symmetric, and transitive. Then for each $x \in X$, recall that Indiff(x) $= \{y \in X : y \sim x\}$. Show that the sets Indiff(x), ranging over $x \in X$, partition the set X. That is, show that each $x \in X$ is in at least one Indiff(y) and that for any pair x and y, either Indiff(x) = Indiff(y) or Indiff(x) $\cap$ Indiff(y) $= \emptyset$.

■ 3. Prove proposition 2.3. (Hint: Use proposition 2.1.)

■ 4. (a) Prove proposition 2.4.

A "problem" with proposition 2.4 is that it assumes that we have the entire function c. If we are trying to test the basic preference-based choice model on observed data, we will have less data than all of c in two respects. First, for sets $A \subseteq X$ where $c(A)$ contains more than one element, we are apt to observe only one of the elements. And we will not see $c(A)$ for all subsets $A \subseteq X$, but only for some of the subsets of X.

(b) Show that the first of these problems is virtually unresolvable. Assume that when we see $x \in A$ chosen from A, this doesn't preclude the possibility that one or more $y \in A$ with $y \neq x$ is just as good as x. Prove that in this case, no data that we see ever contradict the preference-based choice model of section 2.1. (Hint: This is a trick question. If you see the trick, it takes about two lines.)

By virtue of (b), we either will need to assume that we see all of $c(A)$ for those subsets A where we observe choice, or we otherwise need something like local insatiability to tell us when something not chosen is revealed strictly worse than what is chosen.

(c) Suppose that, for some (but not all) subsets $A \subseteq X$, we observe $c(A)$. Show that these partial data about the function c may satisfy Houthakker's axiom and still be inconsistent with the standard preference-based choice model. Formulate a condition on these data such that if the data satisfy this condition, the data are not inconsistent with the standard model. If it helps, you may assume that X is a finite set. (Hint: Proposition 2.12 may help give you the right idea.)

■ 5. Consider the following preferences: $X = [0,1] \times [0,1]$, and $(x_1, x_2) \succ (x_1', x_2')$ if either $x_1 > x_1'$ or if $x_1 = x_1'$ and $x_2 > x_2'$. These are called *lexicographic* preferences, because they work something like alphabetical order; to rank any two objects, the first component (letter) of each is compared, and if those first components agree, the second components are considered. Show that this preference relation is asymmetric and negatively transitive but does not have a numerical representation. (Hint: You should have no problem with the first part of the statement, but the second is more difficult and requires that you know about countable and uncountable sets. This example will appear in almost every book on choice theory, so if you get badly stuck, you can look it up.)

■ 6. Prove proposition 2.9.

■ 7. *Part (a) is not too hard if you know the required mathematics. Part (b) is extremely difficult; see Kannai (1977) if you can't do it and are insatiably curious.*

(a) Consider the following two utility functions defined on the positive orthant in R^2:

$$U_1(x_1, x_2) = \begin{cases} x_1x_2 & \text{if } x_1x_2 < 4, \\ 4 & \text{if } 4 \le x_1x_2 \le 8, \text{ and} \\ x_1x_2 & \text{if } 8 \le x_1x_2. \end{cases}$$

$$U_2(x_1, x_2) = \begin{cases} x_1x_2 & \text{if } x_1x_2 < 4, \\ 4 & \text{if } x_1x_2 = 4 \text{ and } x_1 \ge x_2, \\ 5 & \text{if } x_1x_2 = 4 \text{ and } x_1 < x_2, \text{ and} \\ x_1x_2 + 1 & \text{if } x_1x_2 > 4. \end{cases}$$

Show that the corresponding two preferences are both convex. Show that neither could be represented by a concave utility function. Are either or both of the corresponding preferences semi-strictly convex? Are either or both of the corresponding preferences continuous?

(b) There are preferences that are both semi-strictly convex and continuous but that don't admit any concave numerical representations. One such utility function is

$$U_3(x_1, x_2) = \begin{cases} \frac{2(1+x_2)}{2-x_1} - 1 & \text{if } x_1 + x_2 \le 1, \text{ and} \\ x_1 + x_2 & \text{if } x_1 + x_2 \ge 1. \end{cases}$$

Prove that preferences given by this utility function cannot be represented by any concave function.

■ 8. We assumed local insatiability throughout our discussion of Marshallian demand. One way to obtain this property indirectly is to assume that (a) preferences are semi-strictly convex, (b) preferences are globally insatiable — for all $x \in X$ there exists some $y \in X$ with $y \succ x$, and (c) the consumption set X is convex. Prove that (a), (b), and (c) imply that preferences are locally insatiable.

■ 9. Marshallian demand must be homogeneous of degree zero in prices and income and, as long as preferences are locally insatiable, must satisfy Walras' law with equality. Proposition 2.12 maintains that a finite set of demand data is consistent with maximization of locally insatiable preferences if and only if the data satisfy GARP. Hence satisfaction of GARP must preclude demand that is not homogeneous of degree zero or that fails to satisfy Walras' law with equality. If you like mathematical puzzles, work out why this is. How is it that satisfaction of GARP rules these things out? (Hint: In the definition of revealed preference preceding proposition 2.12, what relation is presumed about the superscripts i and j?)

■ 10. Suppose that in a world with K goods, a locally insatiable consumer's preferences $\succ$ have the following property. Her preferences are strictly convex in the first j goods. That is, if x^1 and x^2 are two distinct bundles of goods with $x^1 \succeq x^2$ and $x_i^1 = x_i^2$ for all $i = j+1, j+2, \ldots, K$, and if $a \in (0,1)$, then $ax^1 + ax^2 \succ x^2$. And she has no use whatsoever for the goods with index above j. If x^1 and x^2 are two bundles of goods with $x_i^1 = x_i^2$ for $i = 1, 2, \ldots, j$, then $x^1 \sim x^2$. Show that if prices are strictly positive, this consumer's solution to (CP) is unique.

■ 11. A particular consumer has Marshallian demand for two commodities given as follows

$$x_1(p_1, p_2, Y) = \frac{Y}{p_1 + 2p_2} \quad \text{and} \quad x_2(p_1, p_2, Y) = \frac{2Y}{p_1 + 2p_2}.$$

This is valid for all price and income levels that are strictly positive. This consumer assures me, by the way, that at prices (p_1, p_2) and income Y, his Marshallian demand is strictly better for him than anything else he can afford.

Does this consumer conform to the model of the consumer used in this chapter? That is, is there some set of preferences, given by a utility function, such that maximization of that utility subject to the budget constraint gives rise to this demand function?

If the answer to the first part of the question is yes, how much about the consumer's utility function and/or preferences can you tell me? Can you nail down the consumer's utility function precisely? (If you answer this yes, and if you've ever played the board game Monopoly, "go directly to jail.") Can you nail down precisely the consumer's preferences over consumption bundles? If not, how much can you say about the consumer's preferences?

■ 12. The purpose of compensated demand functions is to try to isolate the substitution and income effects on demand arising from a change in price. Hicksian compensation holds the consumer to a given level of utility, so that we have the Hicksian demand function, $h(p, u)$, as a function of the level of prices p and the level of utility u to which we wish to hold the consumer. An alternative is *Slutsky* compensation, where we hold the consumer to the income needed to purchase a given bundle of goods. Formally, define $s(p, x)$ as the demand of the consumer at prices p if the consumer is given just enough income to purchase the bundle x at the prices p.

We learned that the matrix of Hicksian substitution terms, whose (i, j)th element is $\partial h_i/\partial p_j$, is symmetric and negative semi-definite. Suppose we looked at the matrix of Slutsky substitution terms, whose (i, j)th entry is $\partial s_i/\partial p_j$. This matrix is also symmetric and negative semi-definite. Prove this. (Hint: Express $s(p, x)$ in terms of the Marshallian demand function. Then see what this tells you about the matrix of Slutsky substitution terms.)

■ 13. In a three-good world, a consumer has the Marshallian demands given in table 2.2. Are these choices consistent with the usual model of a locally insatiable, utility-maximizing consumer?

prices			income	demand		
p_1	p_2	p_3	y	x_1	x_2	x_3
1	1	1	20	10	5	5
3	1	1	20	3	5	6
1	2	2	25	13	3	3
1	1	2	20	15	3	1

Table 2.2 Four values of Marshallian demand.

■ 14. Let Σ denote the substitution matrix of some given Marshallian demand system. That is,

$$\Sigma_{ij} = \frac{\partial x_i(p, Y)}{\partial p_j} + \frac{\partial x_i(p, Y)}{\partial Y} x_j(p, Y).$$

Prove that $p\Sigma = 0$, where p is to be thought of as a $1 \times K$ vector, Σ as a $K \times K$ matrix, and 0 as a $1 \times K$ vector. (Hint: Try showing first that $\Sigma p' = 0$, where p' is the transpose of p. Also, see Euler's law on page 254.)

■ 15. Construct a Marshallian demand function in which one good is a Giffen good. Construct a demand function in which two goods i and j are substitutes but $\partial x_i/\partial p_j < 0$. Be sure that you have given a Marshallian demand function!

chapter three

Choice under uncertainty

Up to this point, we've been thinking of the bundles or objects among which our consumer has been choosing as "sure things" — so many bottles of wine, so many cans of beer, so many shots of whisky. Many important consumption decisions concern choices the consequences of which are uncertain at the time the choice is made. For example, when you choose to buy a car (new or used), you aren't sure what the quality is. When you choose an education, you aren't sure about your abilities, later opportunities, the skills of your instructors, etc. Both in financial and real markets, commodities of risky or uncertain character are traded all the time.

Nothing in the theory up to this point precludes such commodities. A can of Olympia beer is a definite thing — a share of GM is another — and we could simply proceed by taking as given the consumer's preferences for bundles that contain so many cans of beer, so many shares of GM, and so on. But *because* there is special structure to the commodity *a share of GM* (or, rather, because we can model it as having a special structure), we are able to (a) make further assumptions about the nature of our consumer's preferences for such things and, thereby, (b) get out something more concrete about the consumer's demand for these kinds of commodities. This is the basic plot of this chapter.

We begin in section 3.1 with the theory of von Neumann–Morgenstern expected utility. In this theory, uncertain prospects are modeled as probability distributions over a given set of prizes. That is, the probabilities of various prizes are given as part of the description of the object — probabilities are *objective*. Section 3.2 takes up the special case where the prizes in section 3.1 are amounts of money; then one is able to say a bit more about the nature of the utility function that represents preferences. In section 3.3, we briefly discuss a few applications of this theory to the topic of market demand. In section 3.4, we turn to a richer theory, where uncertain prospects are functions from "states of nature" to prizes, and where probabilities arise (if at all) *subjectively*, as part of the representation of a consumer's preferences. In section 3.5 we will explore theoretical and empirical problems with these models. Finally, we will turn briefly to a normative (re)interpretation of this development. While our emphasis is on the use of these models as descriptive models of choice, they are also used as normative guides to aid consumers who must make choices. This is sketched in section 3.6.

3.1. Von Neumann–Morgenstern expected utility

Setup

To begin, think of a basic set of "prizes" X — these are just like our commodity bundles from before — and then a larger set of *probability distributions* over the prizes. Let P denote the set of probability distributions with prizes in X. For the time being, we will assume that P consists only of probability distributions that have a finite number of possible outcomes — such probability distributions are called *simple*. Formally,

Definition. *A* ***simple probability distibution*** *p on X is specified by*

(a) a finite subset of X, called the ***support*** *of p and denoted by* $\mathrm{supp}(p)$, *and*

(b) for each $x \in \mathrm{supp}(p)$ a number $p(x) > 0$, with $\sum_{x \in \mathrm{supp}(p)} p(x) = 1$.

The set of simple probability distributions on X will be denoted by P.

To take an example, suppose that X is the positive orthant in R^2, where $x = (x_1, x_2)$ represents x_1 cans of beer and x_2 bottles of wine. A typical simple probability distribution is one with support $\{(10, 2), (4, 4)\}$, $p((10, 2)) = 1/3$, and $p((4, 4)) = 2/3$. This represents a one-third chance of receiving 10 cans of beer and 2 bottles of wine and a two-thirds chance of getting 4 cans and 4 bottles. We will depict simple probability distributions by *chance nodes*; the example just given is depicted in figure 3.1. Note that the numbers on branches are the probabilities, with the prizes written out at the end of the branches.

Some terminology goes along with this; members x of X will be called *prizes* or *outcomes*. Members p of P will be called *lotteries*, *gambles*,

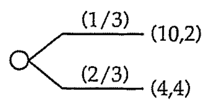

Figure 3.1. A simple probability distribution.

and *probability distributions*, all interchangeably.[a]

And some notation follows: The lottery that gives the prize x with probability one will be written δ_x.

How does this do as a model of the commodity space that we imagine is "out there?" We can say three things here: First, if you thought of commodity bundles as so many cans of beer, so many shares of GM, etc., then you might have thought in terms of a vector of probability distributions instead of a probability distribution on vectors. That is, in terms of our example, we would write a typical commodity bundle as a vector (p_1, p_2) where p_1 is a probability distribution over the number of cans of beer and p_2 is a probability distribution over the number of bottles of wine. But the way we are doing things is superior. You will learn why this is in section 3.3, but for now (if it makes sense) note that if you have a vector of probability distributions, you won't know about any correlations in the prizes they give. A probability distribution over vector prizes tells you not only the marginal distribution on each component but also all conditional and joint probabilities.

Second, the assumption that every distribution has finite support may seem rather limiting. For one thing, we can't in this framework represent a gamble constructed as follows: I flip a coin until the first time it is tails and give you as many dollars as times I flipped the coin, which would give you \$1 with probability 1/2, \$2 with probability 1/4, and so on. Or if the prize space X was R, representing an amount of money, and it was analytically convenient to have Normal probability distributions, our formalism wouldn't be adequate. We will address this shortcoming at the end of this section.

Third, the probabilities all come as part of the description of the object — probabilities are "objective" instead of "subjective." But in real world applications, there may be no objective probability for a random event. For example, suppose an entrepreneur is considering a venture that will earn her a given amount of money if a certain technique for gene splicing works. To use our model in describing her choice problem, we need to know the probability that this technique will work, something about which well-informed individuals might disagree. We'll deal with subjective probabilities in section 3.4.

To complete the setup, we need one more concept. Suppose we have two simple probability distributions p and q and a number α that lies

[a] For readers who know the terminology of probability theory, we tend to use *lotteries* and *gambles* when the term *random variable* is appropriate, while *probability distribution* is used when that is appropriate.

between zero and one, inclusive. Then we can form a new probability distribution, written $\alpha p + (1-\alpha)q$, in two steps:

(a) The support of this new probability distribution is the union of the supports of p and q

(b) If x is some member of this union, then the probability given by $\alpha p + (1-\alpha)q$ to x is $\alpha p(x) + (1-\alpha)q(x)$, where $p(x)$ is understood to be zero if x is not in the support of p, and similarly for q

An example may help. Suppose p gives probabilities .3, .1, and .6 to prizes x, y, and z, respectively, and q gives probabilities .6 and .4 to prizes x and w, respectively. We form, say, $(1/3)p + (2/3)q$ as follows: The support of $(1/3)p + (2/3)q$ is $\{x, y, z, w\}$, and the probabilities it gives to its four possible prizes are, respectively,

for x, $(1/3)(.3) + (2/3)(.6) = .5$
for y, $(1/3)(.1) + (2/3)(0) = .0333\ldots$
for z, $(1/3)(.6) + (2/3)(0) = .2$
for w, $(1/3)(0) + (2/3)(.4) = .26666\ldots$

When we depict probability distributions such as $(1/3)p + (2/3)q$ in the example just given, we sometimes will draw a *compound lottery*. For example, in figure 3.2(a), we show $(1/3)p + (2/3)q$ as a compound lottery: a lottery whose outcomes are the lotteries p and q. In figure 3.2(b), we depict the one-stage lottery to which this compound lottery reduces. *In our setup, if not our pictures, 3.2(a) and (b) depict precisely the same thing.* In a minute, we will assume that we are given some individual's preferences over all simple lotteries, and it is implicit that *if* the individual doesn't regard 3.2(a) and (b) as precisely the same object, then at least she is indifferent between them. As we will discuss in section 3.5, this is somewhat counterfactual. Individuals, faced with choices that involve the sorts of pictures in figure 3.2, sometimes do make different choices depending on whether the "picture" is 3.2(a) or (b). The standard theory that we are about to develop does not permit this.

Axioms for preference

Now assume that our consumer has preferences over the set P of all simple probability distributions on X, given, as before, by a relation $\succ$ that expresses *strict preference*. We insist on two properties immediately.

Assumption 1. $\succ$ must be ***asymmetric and negatively transitive.***

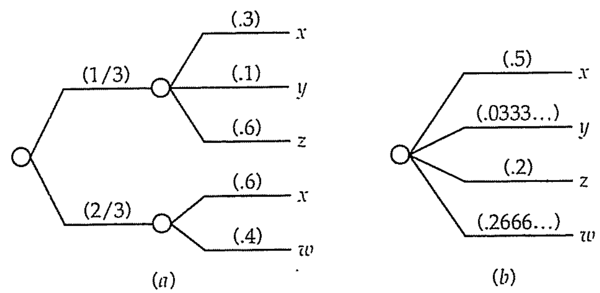

Figure 3.2. Compound and reduced lotteries.
The compound lottery (a) reduces to the single-stage lottery (b) by the laws of probability theory. In our theory, the consumer either *identifies* these two as being the same object or, at least, is indifferent between them.

This is just as before. Also as before, we construct from $\succ$ a *weak preference* relation $\succeq$ and an *indifference* relation $\sim$.

We add to these two properties some properties that exploit the fact that our objects of choice are probability distributions. Consider first

Assumption 2. *Suppose p and q are two probability distributions such that $p \succ q$. Suppose α is a number from the open interval $(0,1)$, and r is some other probability distribution. Then $\alpha p + (1-\alpha)r \succ \alpha q + (1-\alpha)r$.*

This is called the *substitution axiom*. The idea is that in both of the two final probability distributions, the consumer is getting r with probability $1-\alpha$, hence this "same part" won't affect the consumer's preferences. Overall preference depends on how the consumer feels about the differences between the two, that is, on p vs. q. Since we suppose that the consumer prefers p to q, and since $\alpha > 0$ implies that there is some chance that this difference matters, we conclude that the consumer prefers $\alpha p + (1-\alpha)r$ to $\alpha q + (1-\alpha)r$.

Assumption 3. *Suppose that p, q, and r are three probability distributions such that $p \succ q \succ r$. Then numbers α and β exist, both from the open interval $(0,1)$, such that $\alpha p + (1-\alpha)r \succ q \succ \beta p + (1-\beta)r$.*

This is called (for obscure reasons) the *Archimedean axiom*. Think of it this way: Since p is strictly better than q, then no matter how bad r is, we can find some "mixture" of p and r, which we'll write $\alpha p + (1-\alpha)r$, with weight on p close enough to one so this mixture is better than q.

And, similarly, no matter how much better p is than q, we can find a β sufficiently close to zero so $\beta p + (1-\beta)r$ is worse than q. To help you understand this, consider an example in which you might think it is false. Suppose p gives you \$100 for sure, q gives you \$10 for sure, and r consists of your death. You might then say that r is so much worse than q that no probability α however close to one makes $\alpha p + (1-\alpha)r$ better than q. But if you think so, think again. Imagine that you are told that you can have \$10 right now or, if you choose to drive to some nearby location, a check for \$100 is waiting for you. If you are like most people, you will probably get into your car to get the \$100. But this, to a minute degree, increases the chances of your death.

The representation

Does consumer choice conform to these axioms? A lot of experimental evidence suggests that the answer to this is no; see section 3.5 below. Despite this, vast quantities of economic theory are based on the assumption that consumer choice does conform to these axioms, which leads to the following representation of preferences:

Proposition 3.1. *A preference relation $\succ$ on the set P of simple probability distributions on a space X satisfies assumptions 1, 2, and 3 above if and only if there is a function $u : X \to R$ such that*

$$p \succ q \quad \text{if and only if} \quad \sum_{x \in \text{supp}(p)} u(x)p(x) > \sum_{x \in \text{supp}(q)} u(x)q(x).$$

Moreover, if u provides a representation of $\succ$ in this sense, then v does as well if and only if constants $a > 0$ and b exist such that $v(\cdot) \equiv au(\cdot) + b$.

In words, $\succ$ has an *expected utility* representation. Each possible prize has a corresponding utility level, and the value of a probability distribution is measured by the expected level of utility that it provides. Moreover, this utility function is unique up to a positive affine transformation (which is a fancy way of putting the last statement).

This is sometimes called a von Neumann–Morgenstern expected utility representation, since one of the original modern developments of this theory appears in von Neumann and Morgenstern's *Theory of Games and Economic Behavior*. But the form goes back a good deal further, to the eighteenth century and Daniel Bernoulli.

Note that the proposition establishes the existence of a numerical representation for preferences on p. That is, there is a function $U : P \to R$ such that $p \succ q$ if and only if $U(p) > U(q)$. The proposition establishes a good deal more than this. It also establishes that we can assume that this function U takes the form of expected utility of prizes: $U(p) = \sum_{x \in \text{supp}(p)} u(x)p(x)$, for some $u : X \to R$.

Numerical representations, you will recall, are unique up to strictly increasing rescalings. That is, if $U : X \to R$ gives a numerical representation of $\succ$, then so will $V(\cdot) \equiv f(U(\cdot))$ for any strictly increasing f. But if U has the form of expected utility, so that $U(p) = \sum_{x \in \text{supp}(p)} u(x)p(x)$, for some $u : X \to R$, and if f is an arbitrary increasing function, then the function V that results from composing U with f may not have the form of expected utility. You can be sure that V will be another expected utility representation if and only if f is an increasing affine function, or $f(r) = ar + b$ for $a > 0$, in which case the $v : X \to R$ that goes with V will be $v(\cdot) \equiv au(\cdot) + b$.

Continuing on this general point, recall that we said in the previous chapter that there is no cardinal significance in utility differences. That is, if $U(x) - U(x'') = 2(U(x') - U(x'')) > 0$ (where U is defined on X), this doesn't mean anything like x is twice better than x'' than is x'. It just means that $x \succ x' \succ x''$. But in the context of expected utility, if $u(x) - u(x'') = 2(u(x') - u(x'')) > 0$ where u gives an expected utility representation on P, this has cardinal significance. It means that a lottery that gives either x or x'', each with probability 1/2, is indifferent to x' for sure.

How is proposition 3.1 proven? We will sketch the proof here; if you are ambitious and know a bit of real analysis, problem 1 will give you a few hints on how to fill in the details. (It isn't very hard.) First, we'll add one assumption (to be taken away in problem 1): In X are a best prize b and a worst prize w; the consumer in question at least weakly prefers b for sure to any other probability distribution over X, and any other probability distribution over X is a least as good as w for sure. If the consumer is indifferent between b for sure and w for sure, then the representation is trivial, so we'll assume as well that $\delta_b \succ \delta_w$. Now we can use assumptions 1, 2, and 3 to obtain three lemmas.

Lemma 1. *For any numbers α and β, both from the interval $[0,1]$, $\alpha\delta_b + (1-\alpha)\delta_w \succ \beta\delta_b + (1-\beta)\delta_w$ if and only if $\alpha > \beta$.*

Or, in words, if we look at lotteries involving only the two prizes b and w, the consumer always (strictly) prefers a higher probability of winning the better prize.

Lemma 2. *For any lottery $p \in P$, there is a number $\alpha \in [0,1]$ such that $p \sim \alpha\delta_b + (1-\alpha)\delta_w$.*

This result, which is sometimes simply assumed, is called the *calibration* property. It says that we can *calibrate* the consumer's preference for any lottery in terms of a lottery that involves only the best and worst prizes. Note that by virtue of lemma 1, we know that there is exactly one value α that will do in lemma 2; if p were indifferent to two different mixtures of the best and worst prizes, it would be indifferent to two things, one of which is strictly preferred to the other.

Lemma 3. *If $p \sim q$, r is any third lottery, and α is any number from the closed interval $[0,1]$, then $\alpha p + (1-\alpha)r \sim \alpha q + (1-\alpha)r$.*

This is just like the substitution axiom, except that $\succ$ is replaced by $\sim$ here. This is sometimes assumed as an axiom, and it itself is then sometimes called the substitution axiom.

The rest is very easy. For every prize x, define $u(x)$ as that number between zero and one (inclusive) such that

$$\delta_x \sim u(x)\delta_b + (1-u(x))\delta_w.$$

This number $u(x)$ will be the utility of the prize x. We know that one such number exists by lemma 2, and this number is unique by lemma 1. Take any lottery p:

Lemma 4. *For $u : X \to R$ defined as above, any lottery p is indifferent to the lottery that gives prize b with probability $\sum u(x)p(x)$ and w with the complementary probability, where the sum is over all x in the support of p.*

Once we have this result, we can use lemma 1 to finish the main part of the proof: Compare any two lotteries, p and q. The lottery p is indifferent to the lottery that gives b with probability $\sum u(x)p(x)$ and w with the complementary probability, and q is indifferent to the lottery that gives b with probability $\sum u(x)q(x)$ and w with the complementary probability. We know by lemma 1 how to compare the two lotteries over b and w: Whichever gives a higher probability of b is better. But this is precisely the representation.

Proving lemma 4 is fairly simple in concept but rather cumbersome notationally. The idea is that we take each prize x in the support of p and substitute for it the lottery that gives b with probability $u(x)$ and w with probability $1-u(x)$. By lemma 3, each time we make such a substitution we will have a new lottery that is indifferent to p. And when we are done with all these substitutions, we will have the lottery that gives prize b with probability $\sum u(x)p(x)$ and w with probability $1-\sum u(x)p(x)$. Figure 3.3 shows how this works for a particular lottery: We begin with p, which has prizes x, x', and x'' with probabilities $1/2$, $1/3$, and $1/6$, respectively. This is shown in 3.3(a). In 3.3(b) we replace the prize x'' with a lottery giving prize b with probability $u(x'')$ and w with probability $1-u(x'')$; this is the lottery p_1, which we show both as a "compound" lottery and as a "one-stage" lottery. We claim that $p \sim p_1$ by lemma 3: Write p as $(5/6)[(3/5)\delta_x + (2/5)\delta_{x'}] + (1/6)\delta_{x''}$; by lemma 3 this is indifferent to $(5/6)[(3/5)\delta_x + (2/5)\delta_{x'}] + (1/6)[u(x'')\delta_b + (1-u(x''))\delta_w]$, which is p_1. Continuing in this fashion, from p_1 we create p_2, which substitutes for x' in p_1 the lottery that gives b with probability $u(x')$ and w with probability $1-u(x')$; we know that $p_1 \sim p_2$. And then we create p_3 from p_2, substituting for x in p_2. Once again, lemma 3 tells us that $p_2 \sim p_3$. But then $p \sim p_1 \sim p_2 \sim p_3$, and by transitivity of indifference, $p \sim p_3$, which (for the lottery p) is just what lemma 4 claims.

Of course, this is only a demonstration of lemma 4 on a particular example; it is not a proof. The exact proof is left as an exercise (problem 1(d)); if you try this, you should use induction on the size of the support of p.

All this was based on our added assumption that there was a best and worst prize. But this assumption can be done away with; you can either consult one of the standard reference books or try your hand at problem 1(e).

To complete the proof of the proposition, we need to show that if preferences have an expected utility numerical representation, then those preferences satisfy assumptions 1 through 3, and we need to show that a representing utility function u is unique up to positive affine transformations. (That is, any other function v that gives an expected utility representation for fixed preferences $\succ$ satisfies $v \equiv au + b$ for constants $a > 0$ and b.) These parts of the proof are left to you.

To summarize: We began by taking a choice space that had some special structure — a set of probability distributions. We used that structure to pose some axioms for the consumer's preferences that exploited this structure. Then, using those axioms, we showed that a numerical representation for consumer preferences can be created that exploits this structure; viz., the representing function U on P takes the form of expected utility for some function u defined on X. We continue this in the next section, by making further assumptions about X and $\succ$.

But first we address a concern voiced earlier: In this development we only got the representation for probability distributions with finite support. Can this sort of representation be extended to probability distributions such as, say, the Normal distribution? Yes, it certainly can be. We sketch one

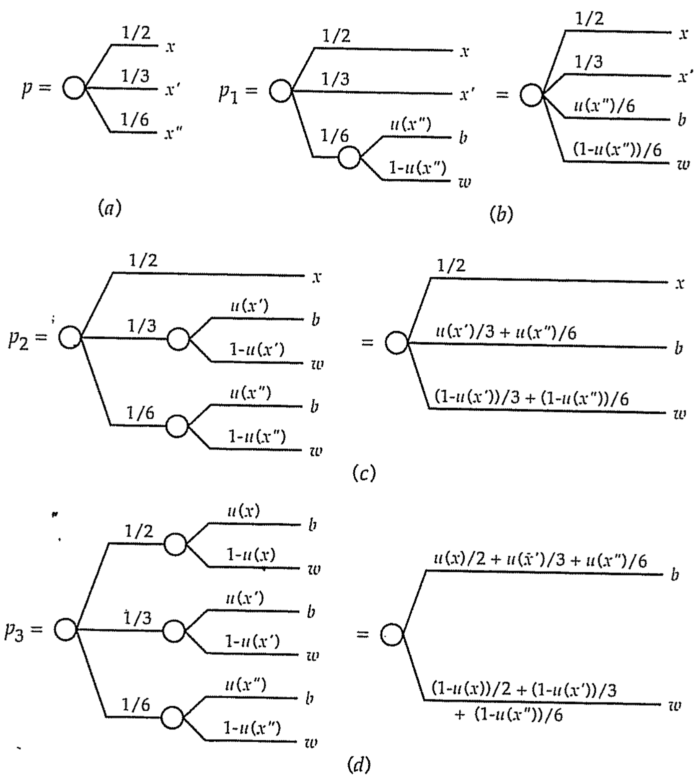

Figure 3.3. Lemma 3 in action.

In the figure above, we successively replace each prize in the lottery p with the lottery between b and w that is just indifferent to that prize. Lemma 3 ensures that, at each step, the lottery replaced is indifferent to the lottery that replaces it. So, in the end, we have a lottery between b and w that is just indifferent to the original lottery. This final lottery gives prize b with probability $\sum_x u(x)p(x)$, and since (by lemma 1) we know that lotteries over b and w are ranked by the probability with which they give b, we know that all lotteries are ranked by their expected utility.

way to proceed, which requires that you know some advanced probability theory. Assume that X is R^k or the positive orthant of R^k. (If you know enough math, take X to be a compact subset of a complete separable metric space.) Let P be the space of all Borel measures on X and assume that preferences are continuous in the weak topology on measures. If assumptions 1, 2, and 3 hold for all probability distributions in P, they hold for the subspace of simple probability distributions, so we produce an expected utility representation for some function $u : X \to R$ that works for simple probability distributions. We can then use the continuity assumption to show that u must be bounded. (If u were unbounded [say] above, we could find a sequence of prizes $\{x_n\}$ with $u(x_n) > 2^n$. Construct the lottery that gives prize x_n with probability $1/2^n$. This has "expected utility" infinity. If you are clever, you should be able to show that this poses real problems for the Archimedean axiom and weak continuity.) And then use the fact that simple probability distributions are dense in the Borel probabilities in the weak topology to extend the expected utility representation to all of P. You can go on to show that u must be continuous (on X) as well as bounded.

The boundedness of u is somewhat limiting in applications. For example, for reasons to be explored later, a very nice utility function in applications is exponential, or $u(x) = -e^{-\lambda x}$, where X is the real line. This utility function is unbounded below. So we would want a way to obtain an expected utility representation for more than simple probability distributions that didn't require bounded u. This can be done as well. Essentially, we obtained bounded u because we took P to be all Borel probability distributions and we assume continuity over all of P. If one restricts P, say to distributions that meet certain tail conditions, then the set of compatible utility functions becomes those u that don't grow "too quickly" relative to the assumed rate at which tail probabilities must die off. For further details, see Fishburn (1970) or Kreps (1988).

3.2. On utility for money

Now we specialize even further to the case where the prizes are dollar values. That is, X is the real line, or some interval thereof. We'll continue to refer to the probability distributions p as lotteries, and we'll continue to restrict attention to simple probability distributions. Throughout, we assume that we have the three assumptions of the previous section and, therefore, an expected utility representation.

In this case, it seems reasonable to suppose that our consumer prefers more money to less. This has a straightforward consequence for the representation, which you should have no difficulty proving.

Proposition 3.2. *Suppose that for all x and y in X such that $x > y$, $\delta_x \succ \delta_y$. This is so if and only if the function u is strictly increasing.*